Remote
Expeditionary
Multi-platform
Operations and
Reconnaissance
Asset

2.014 Final Presentation

Agenda

- Why do we need REMORA?
- What is REMORA?
- How does REMORA work?
- REMORA Design Overview
  - ROV System Design
  - Drone System Design
  - Integrated Boat System
- Next Steps / In-Person Demo

thesis

REMORA is a capable system because

1 robust design

2 testing has proven

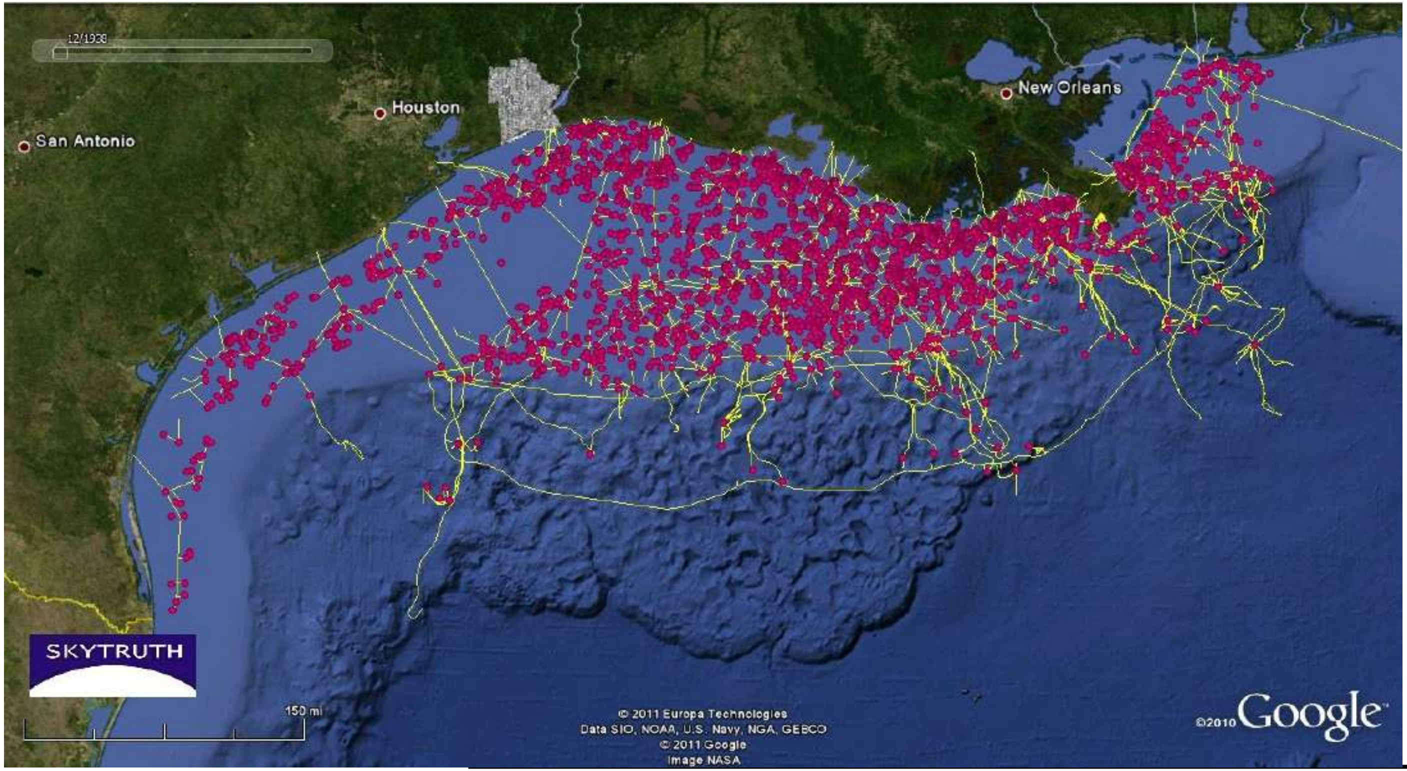
3 future path is clear

Why do we need REMORA?

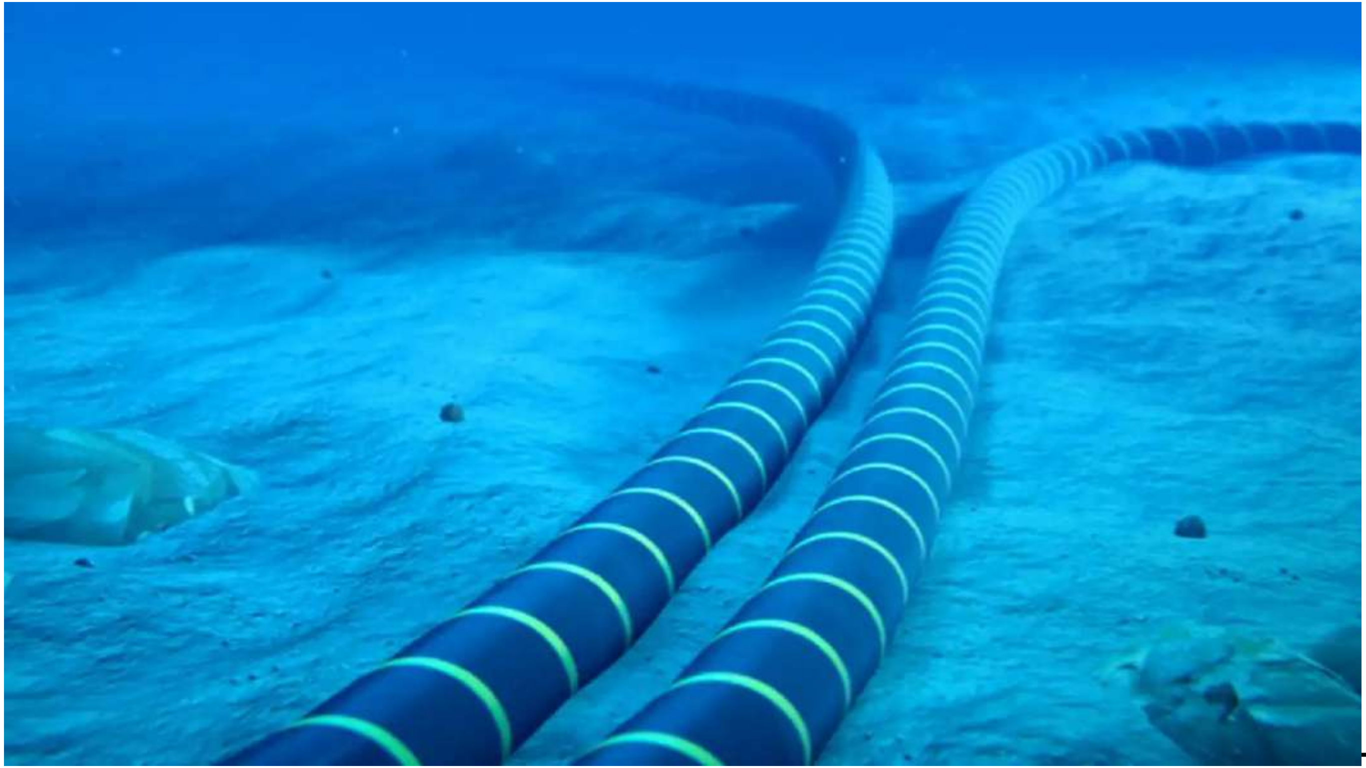
There are currently >15,000 offshore wind turbines

30% of the world's oil/gas is produced offshore

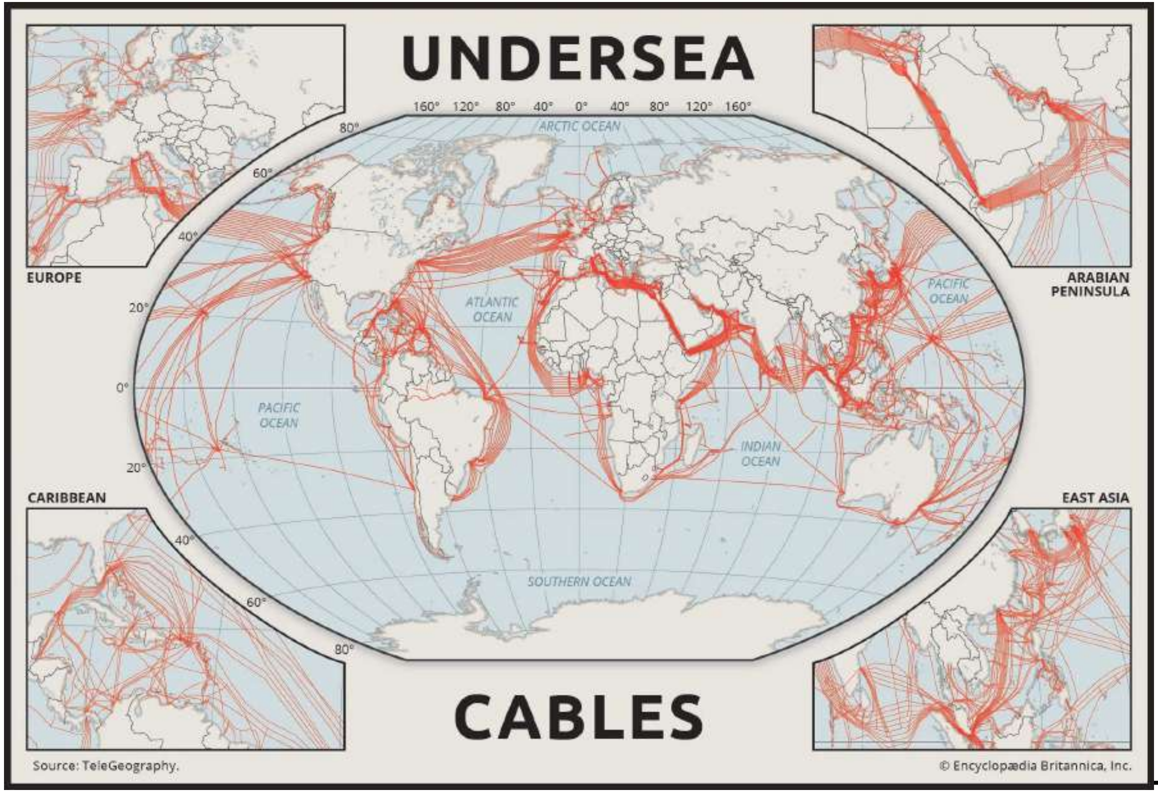
Over 4,000 oil rigs in the Gulf of Mexico alone



95% of the world's data flows through undersea cables



These cables connect our entire world



Marine environments rapidly degrade these structures

Without regular inspections these structures break

Without regular inspections these structures break

400 ft = 30 story building!

Without regular inspections these structures break

Vineyard Offshore Incident (Massachusetts)

- **$853 Million** Damages
- **$10.5 M** Settlements

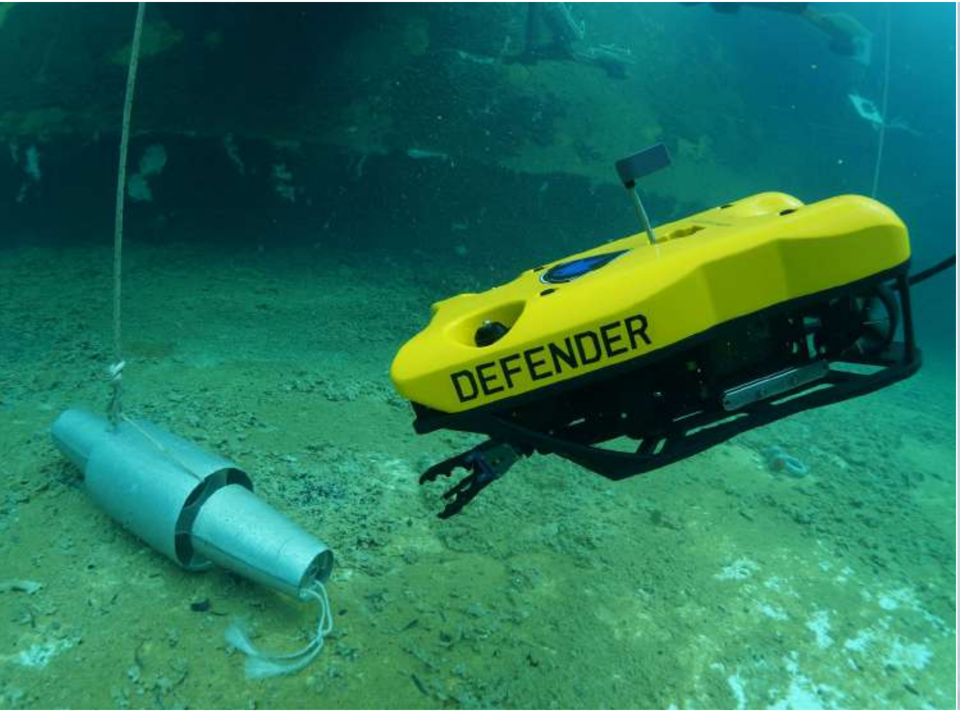
Without regular inspections they become dangerous

Piper Alpha

- **167 dead**
- **$1.7 billion in Losses**

Inspecting offshore assets is costly, dangerous, and slow

Single System Drones/ROVs help, but are still costly and slow



- Inspection is **costly, slow, and dangerous** due to weather and piracy.
- Personnel must be moved up to **200 miles offshore, via helicopter** ($8k / hr) **or boat** ($10k / day).
- **≥12,000** offshore oil platforms worldwide
- Potential **Defense Interests** for **Reconnaissance & Mapping**

Simple aerial turbine inspection: >$15k /day

- Vessel/Helicopter Day Rates Costs: $4,500
- Labor Costs: $10,000
- Overhead costs: $1,000
- Sensor & Inspection Costs: $1,500

Underwater Turbine Inspections: >$20k/day

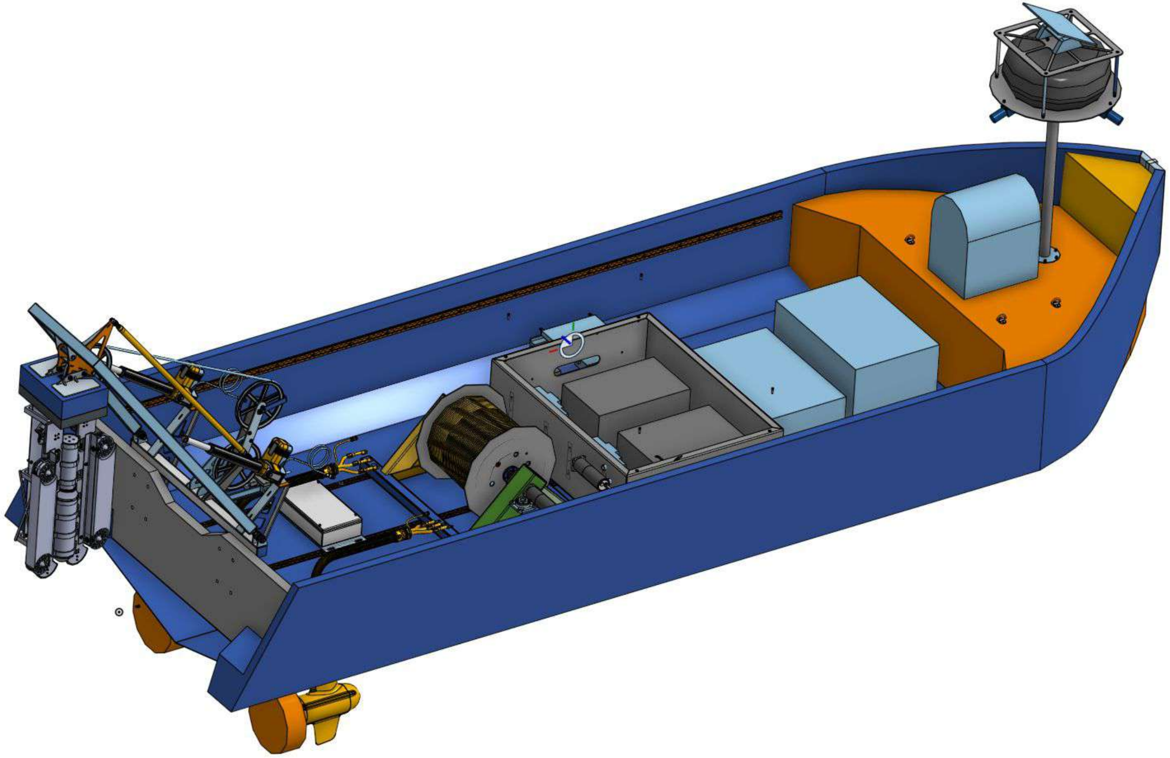
- Vessel/Helicopter Day Rates Costs: $2,500
- Labor Costs: $12,000
- Overhead costs: $1,000
- Sensor & Inspection Costs: $1,500

Our Solution

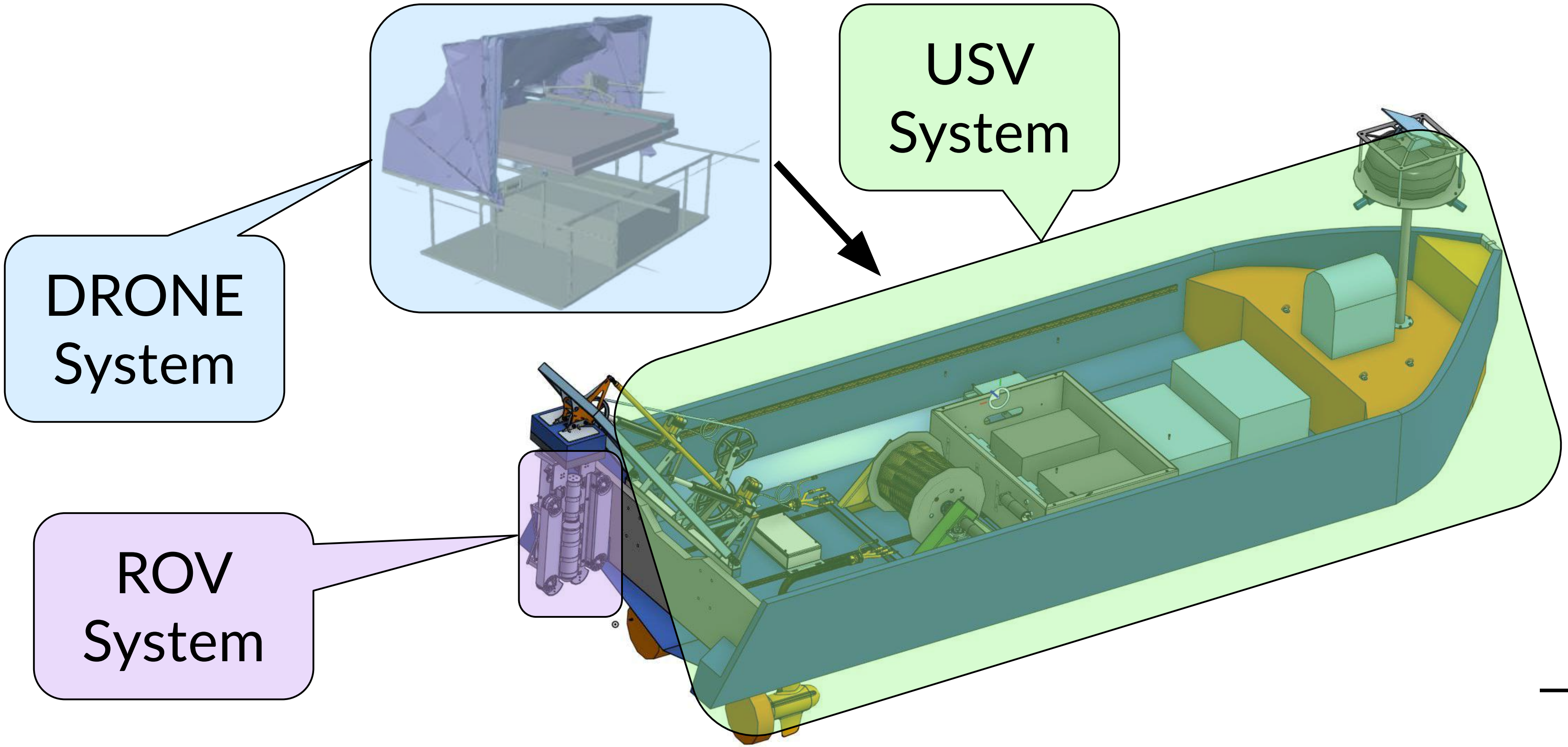
With REMORA, we aim to provide a **80% reduction in offshore asset inspection costs,** eliminating the need for humans to go offshore

What is REMORA?

- Cost Effective
- Time Efficient
- Self-Sustained
- Unmanned Systems
- Conducts routine aerial & underwater surface sweeps for offshore asset inspections



REMORA Consists of 3 Main Systems



Offshore Wind Turbines provide a Market Opportunity for REMORA

- **~56 Billion**, and offshore production capacity is expected to **grow 5x** by 2034[1,2] Fernandez, L. 2024
- **>15,000 current offshore wind turbines**
- **>$20,000/day** to use manned vessels for inspection.[3] DOE, 2023
- **Cost-effective, reliable, unmanned**

REMORA's ROI shows Significant Economic Viability

Manned Inspection Systems:

- Vessel Day Rates Costs
- Labor Costs
- Overhead costs
- Sensor & Inspection Costs
- Cost For Company: **~$17,000**
- Cost To Customer: **~$20,000**
- Profit Margin: **15%**
- Availability based on 2x a Day, 12 on/12 off schedule, weather/personel permitting

Unmanned Inspection Systems:

- Minimal Labor Costs
- Overhead Costs
- Sensor & Inspection Costs
- Cost For Company: **~$2,900**
- Cost to Customer: **~$6,000**
- Profit Margin: **50%**
- **Available 24/7**, weather permitting

REMORA's ROI shows Significant Economic Viability

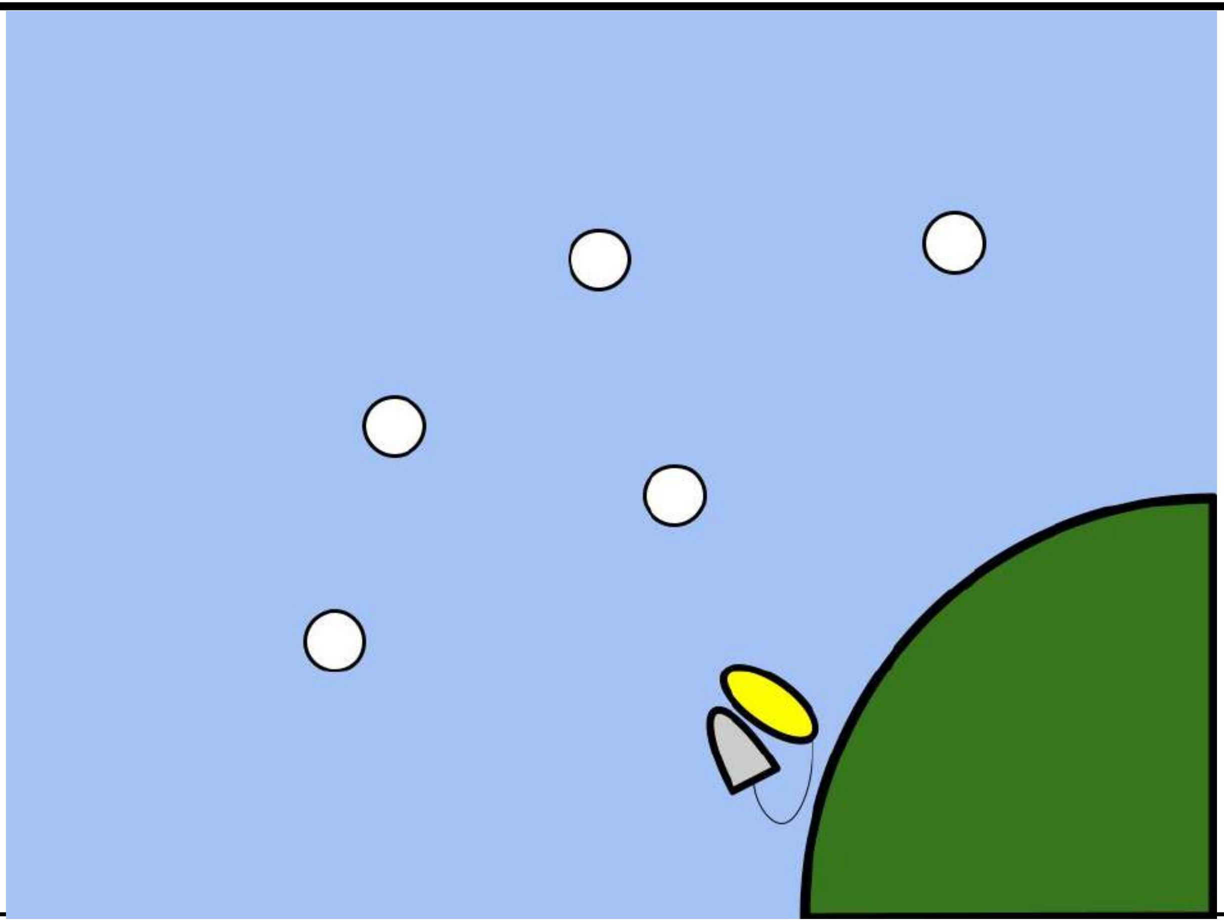
Method	Vessel Day Rate	Labor Costs (8 hr/Day), $25/hr	Sensor and Inspection Costs	Indirect Costs	Total Cost for us ($/day)	Cost to Customer ($/day)	Profit Margin	Deploy Availability
Manned[1]	$2,500	$12,000	$1,500	$1,000	$17,000	**$20,000+**	**15%**	2x a Day, 12 on/12 off schedule, weather/personel permitting
Unmanned	N/A	$400	$1,500	$1,000	**$2,900**	**$6,000**	**>50%**	**24/7, weather permitting**

1: O. Khalid, G. Hao, H. MacDonald, A. Cooperman, F. Devoy McAuliffe, and C. Desmond, "Cost-benefit assessment framework for robotics-driven inspection of floating offshore wind farms," *Wind Energy*, vol. 27, no. 2, pp. 152–164, Feb. 2024, doi: 10.1002/we.2881.

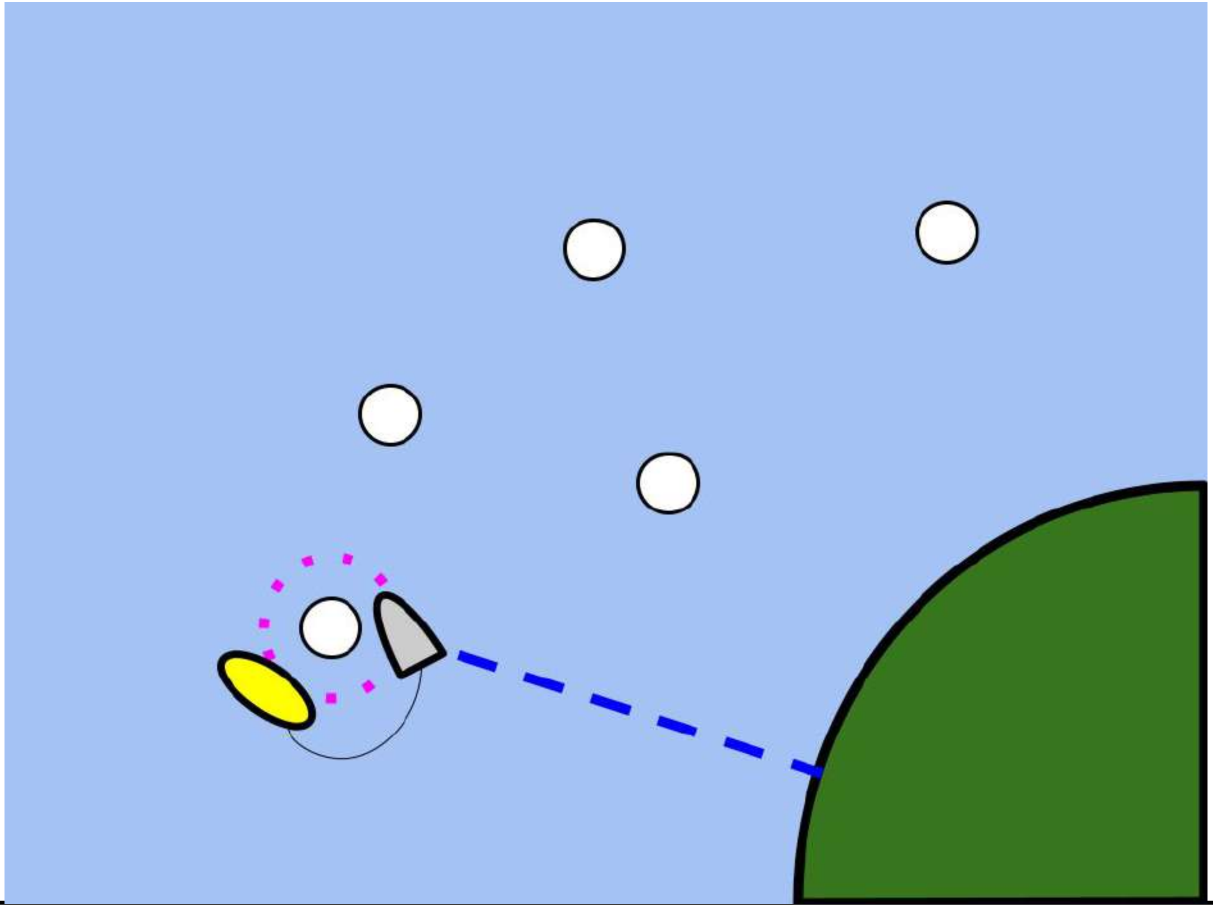
How does REMORA Work?

ROV/DRONE Example Mission

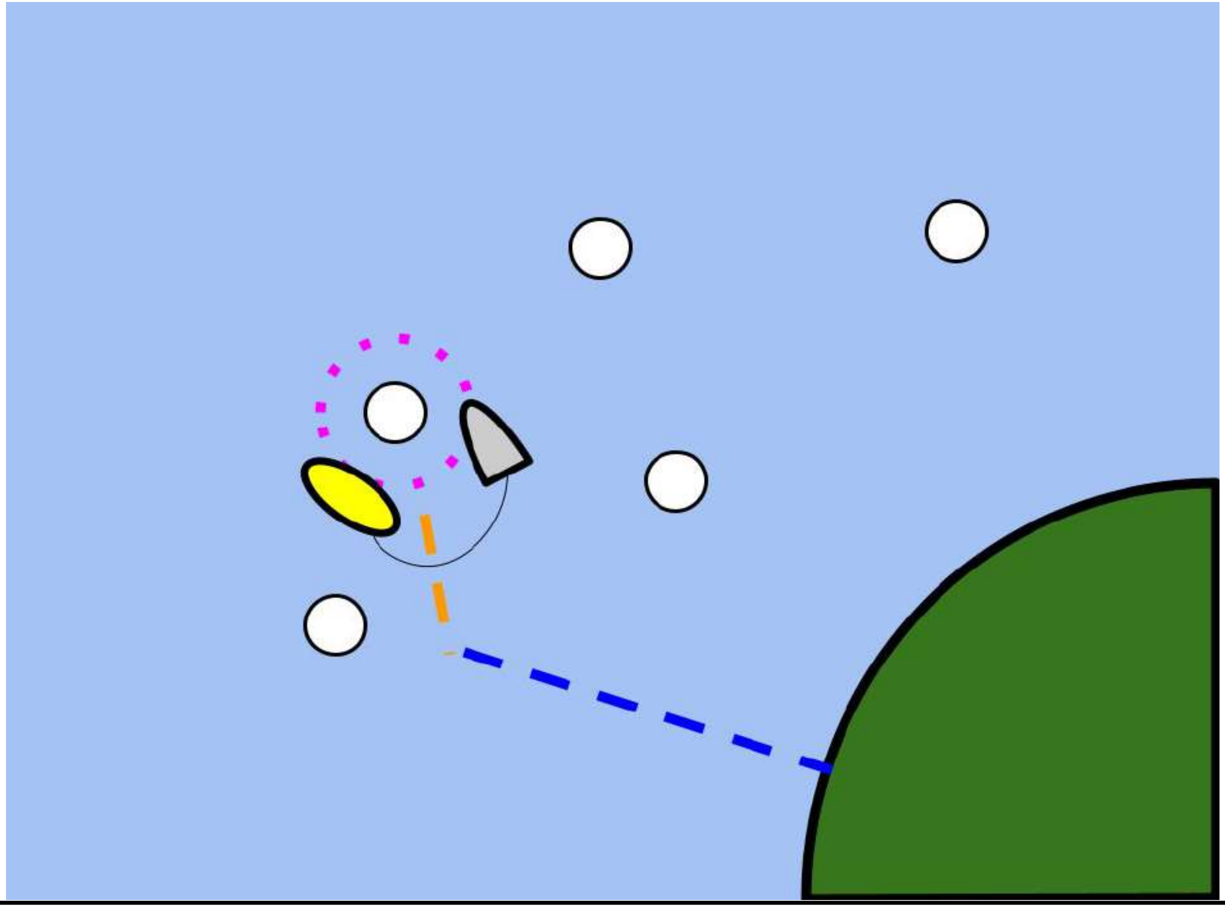
- Offshore Wind Monopiles
- ROV/DRONE
- USV
- USV Shore Deployment/Return
- ROV Mission Operation
- USV Pile to Pile Navigation



ROV/DRONE Example Mission



ROV/DRONE Example Mission



ROV/DRONE Example Mission

Offshore Wind Monopiles

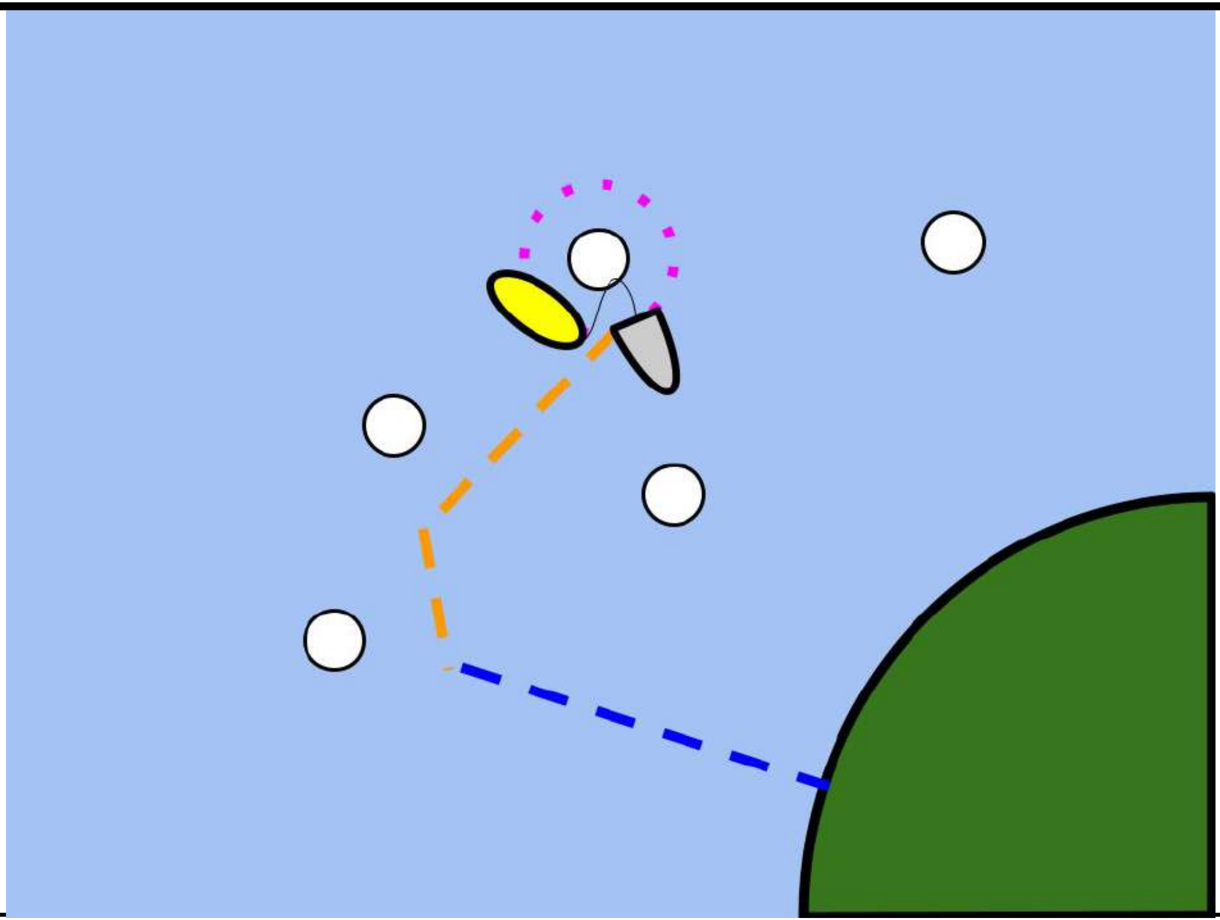
ROV/DRONE

USV

USV Shore Deployment/Return

ROV Mission Operation

USV Pile to Pile Navigation



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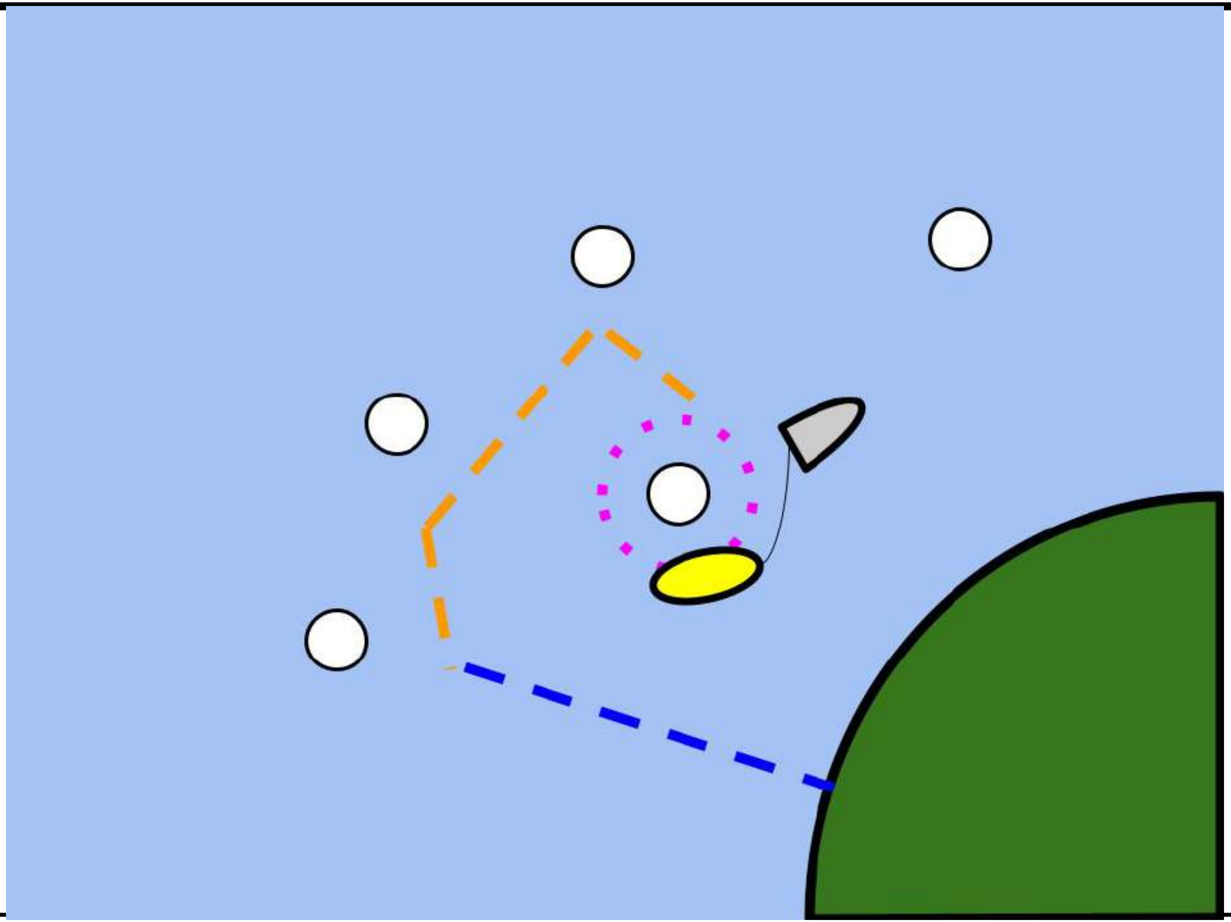
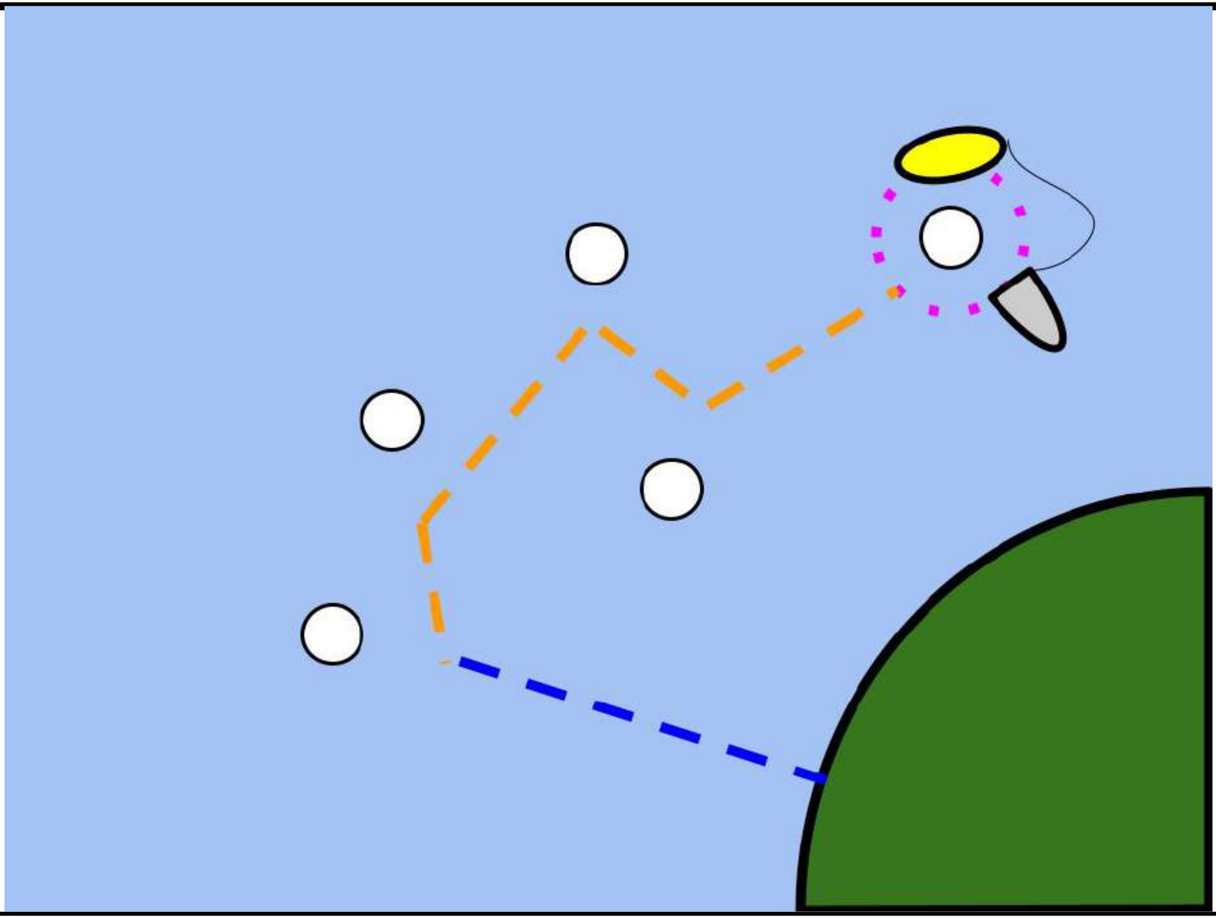


ROV/DRONE Example Mission



ROV/DRONE Example Mission

Offshore Wind Monopiles

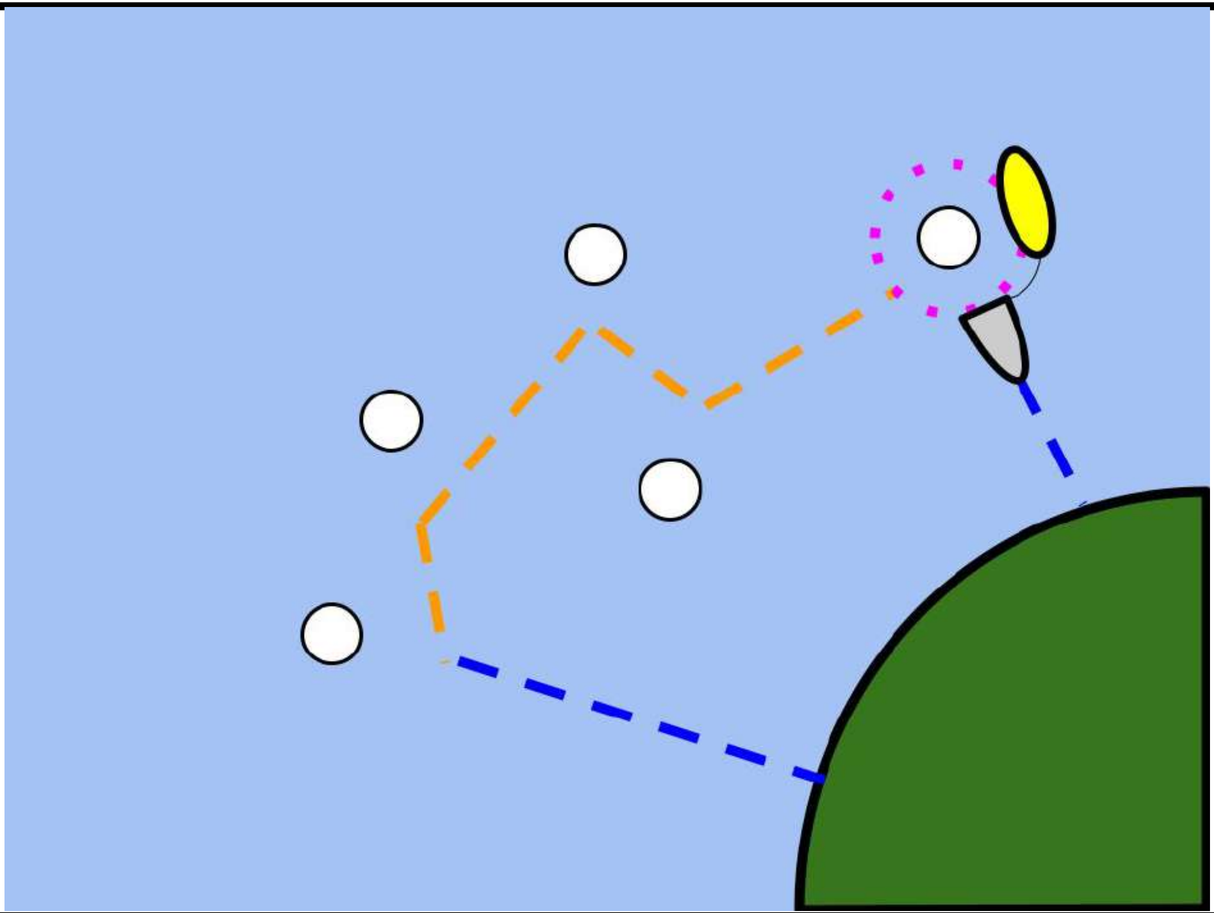
ROV/DRONE

USV

USV Shore Deployment/Return

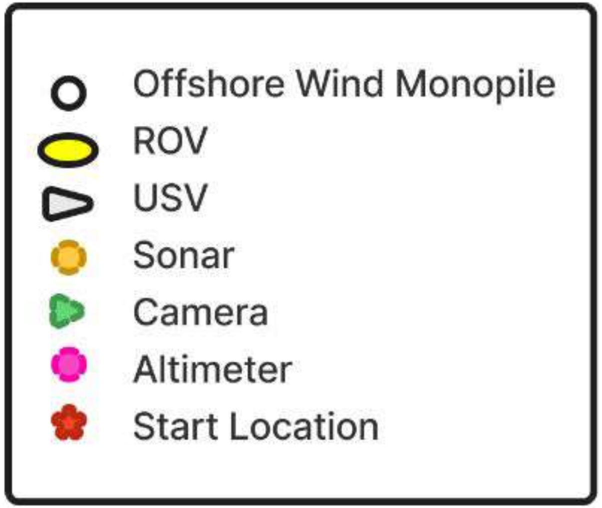
ROV Mission Operation

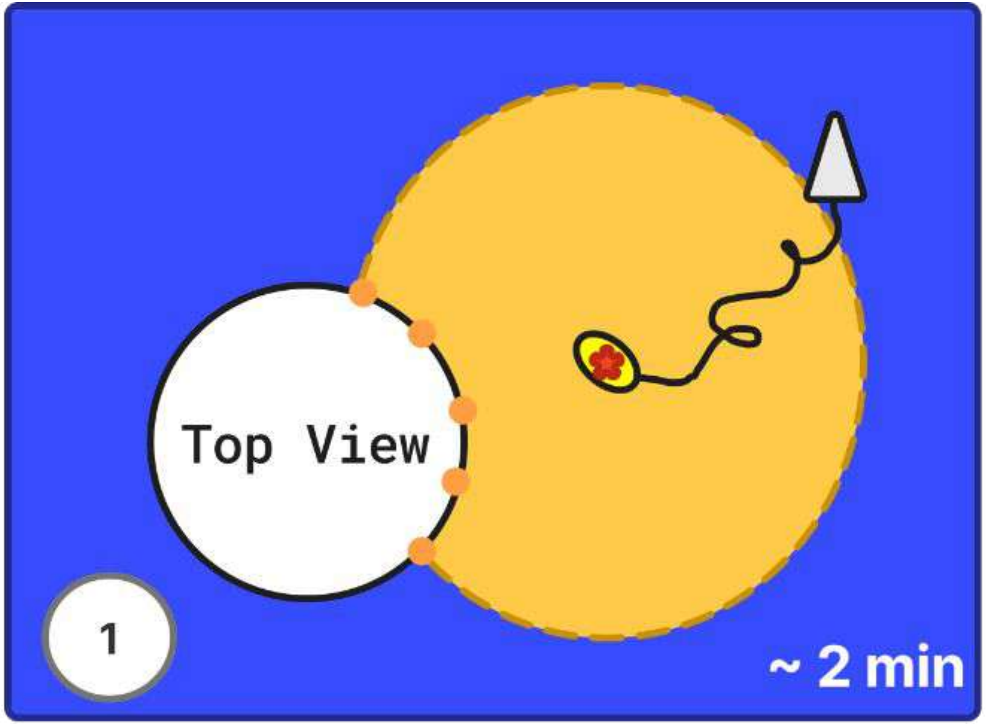
USV Pile to Pile Navigation



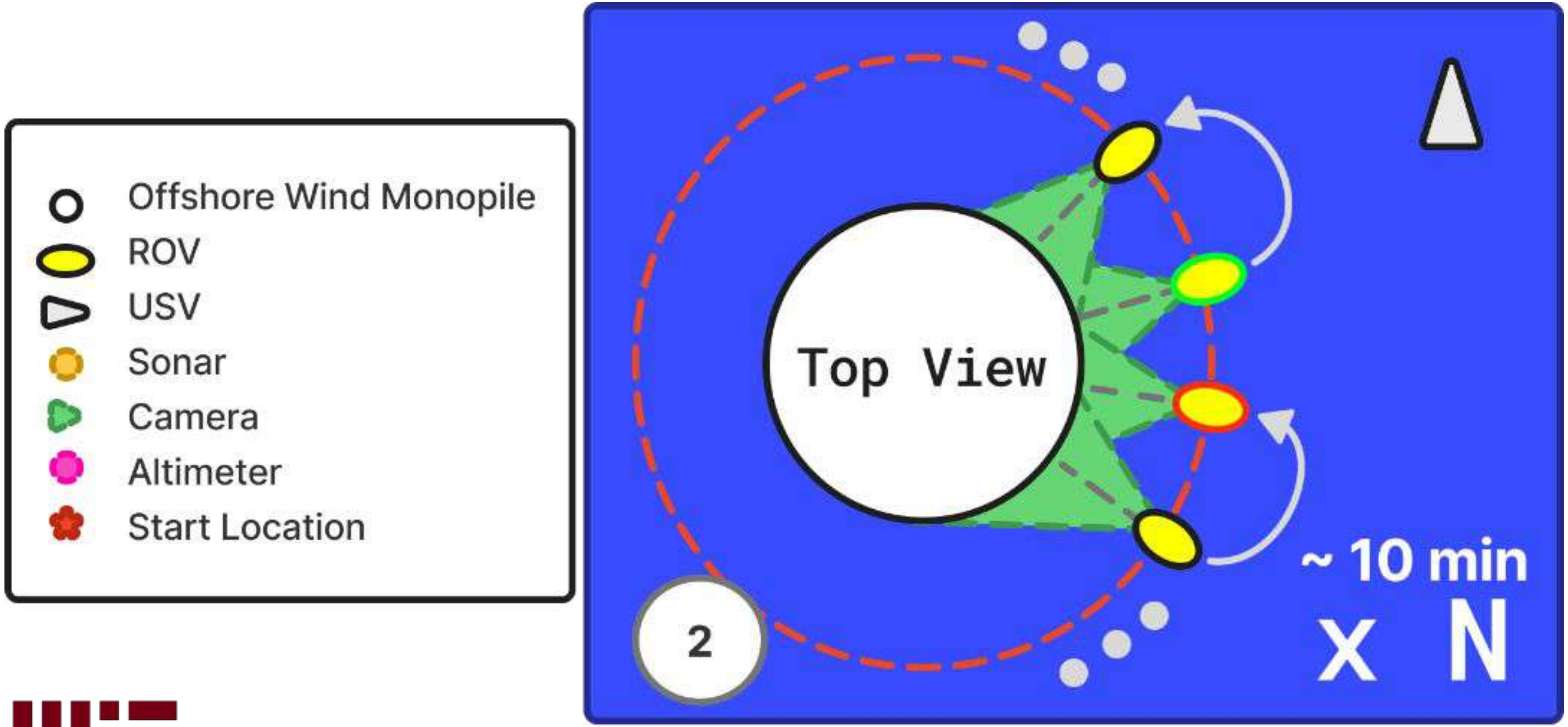
How does REMORA Deployment Work?

Inspection begins with Sonar Identification of Asset

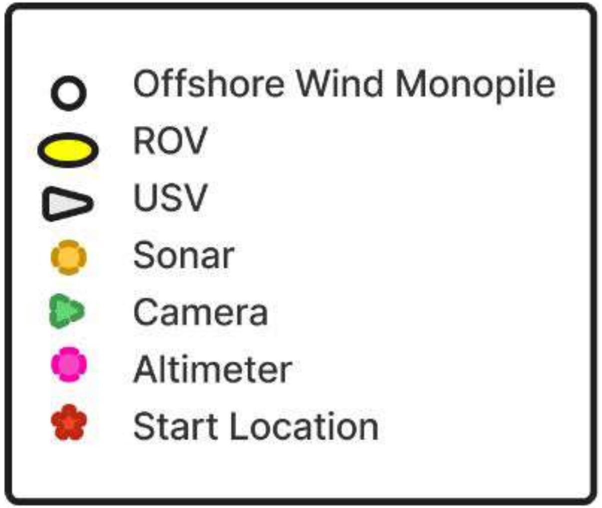


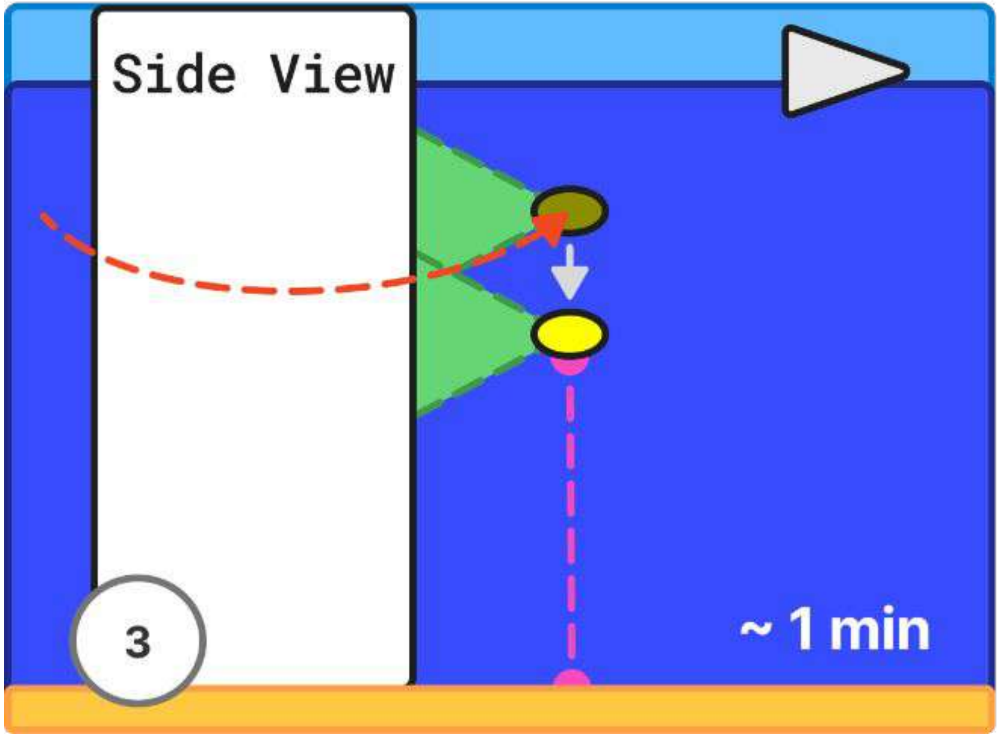


Once Located, the ROV Rotates around Capturing Images

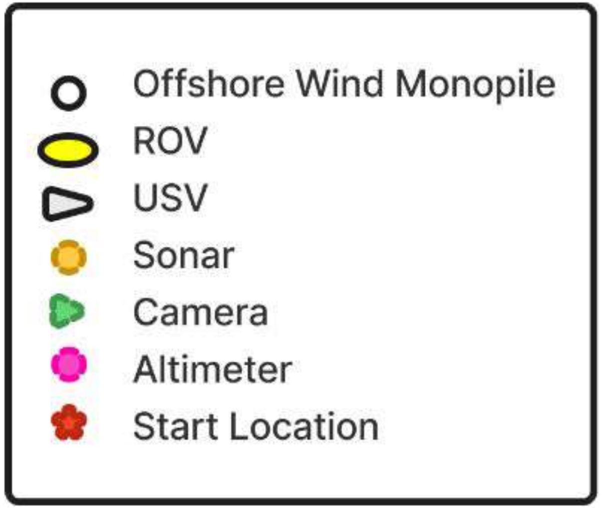


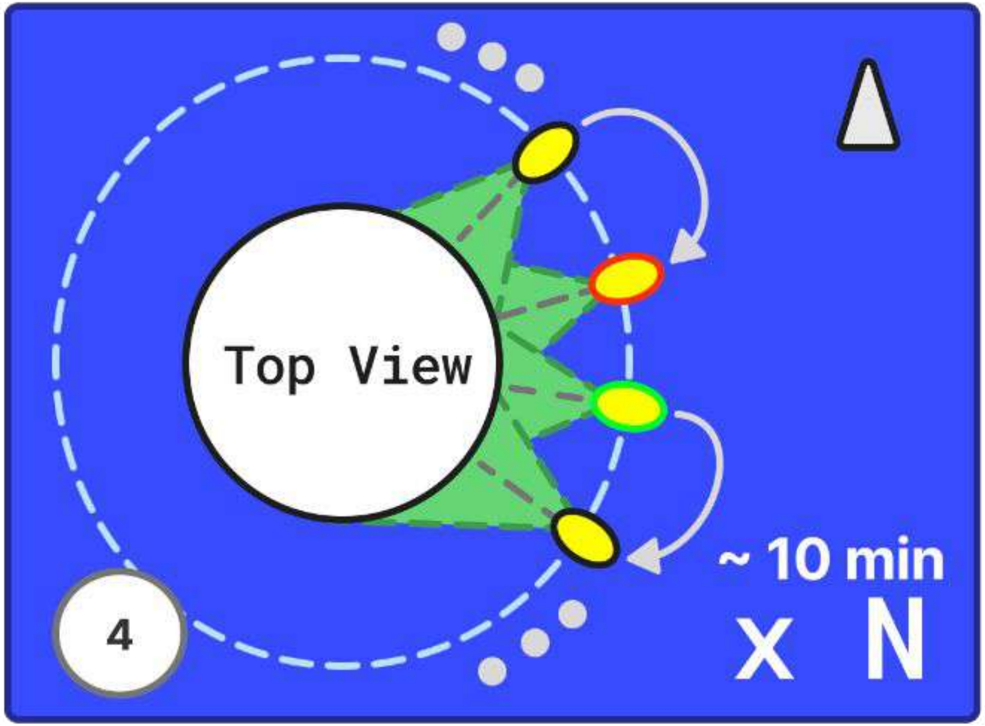
Once Complete, ROV moves to the Next Depth





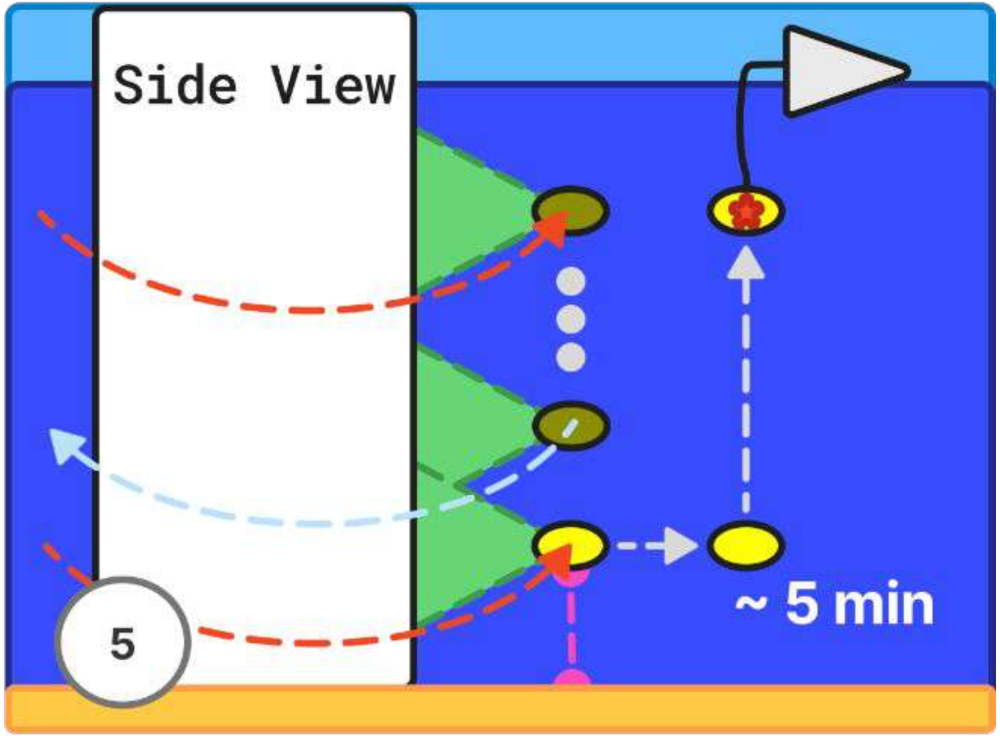
At New Depth the ROV changes its rotation from CW to CCW



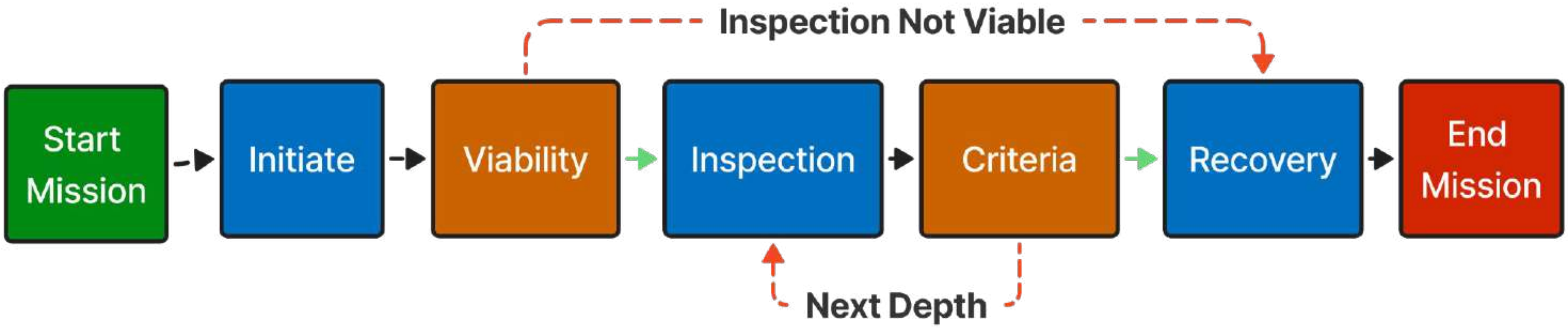


Process is Repeated until the Mission is Completed

Offshore Wind Monopile
ROV
USV
Sonar
Camera
Altimeter
Start Location



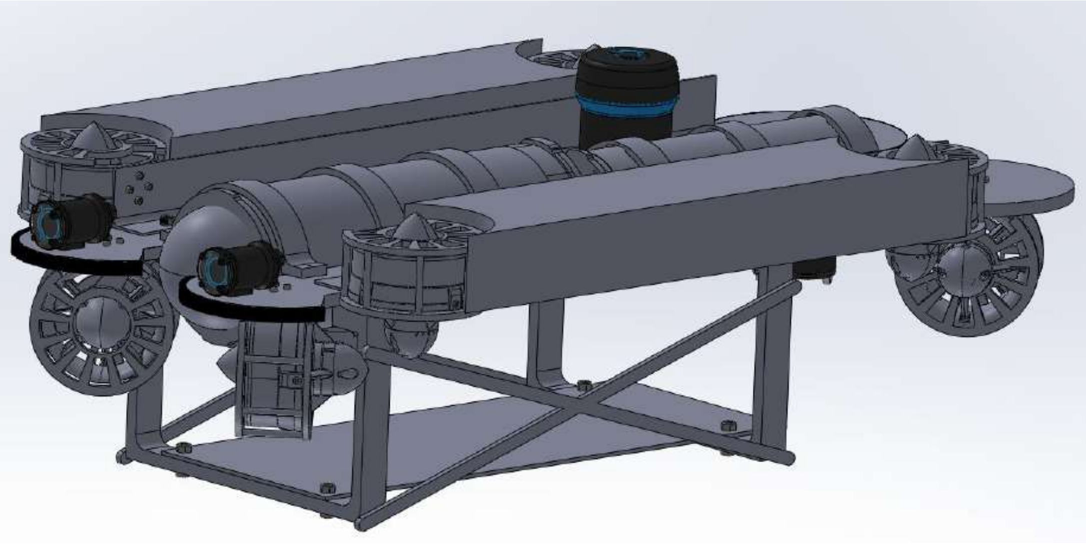
Offshore Asset/Monopile Inspection Flowchart



REMORA Design Overview - ROV System

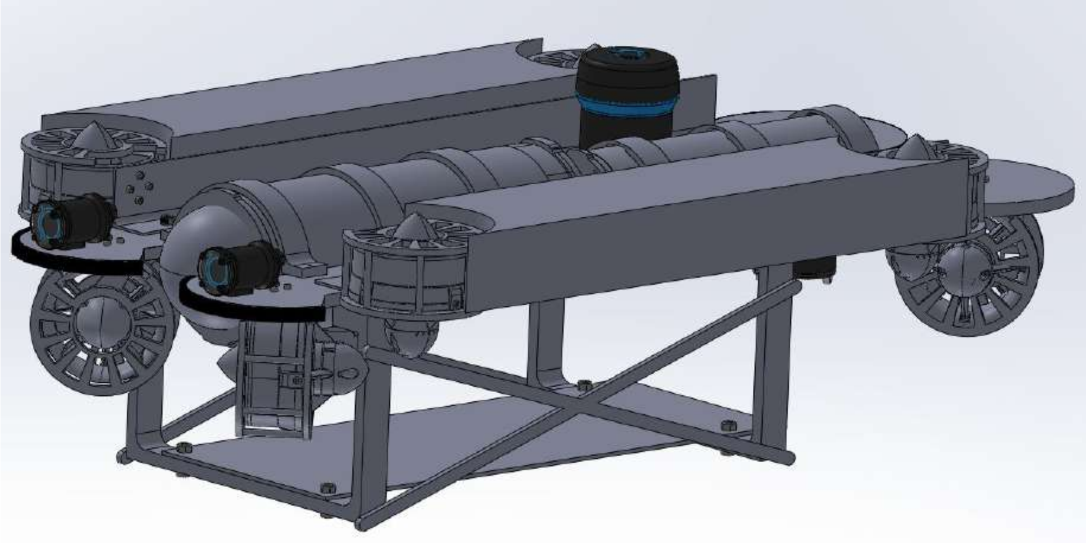
Final Design - Overview + Structure

- BOM Cost: $11,479
- Total Weight: ~15 kg
- Original BlueROV2 Heavy components + modified chassis

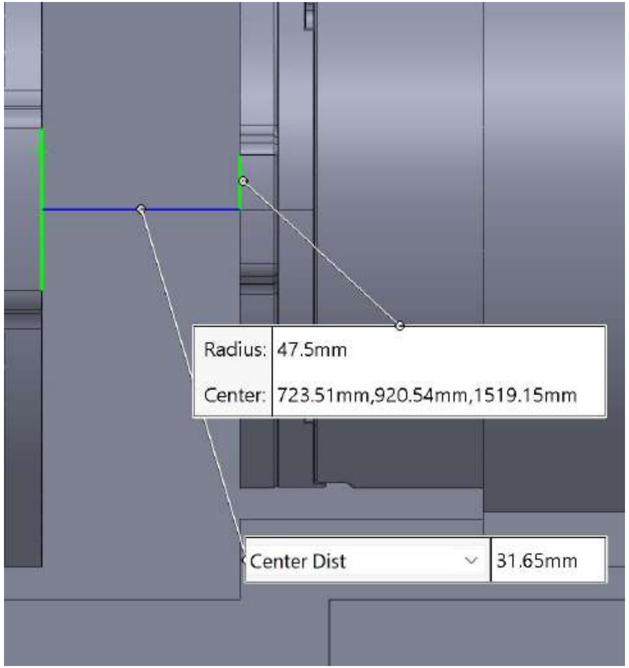


Final Design - Electronics + Sensors

- Paide tether connection
- Testing purposes: 7200mAh battery in rear enclosure
- Two front-facing lights
- Ping360 Sonar + altimeter

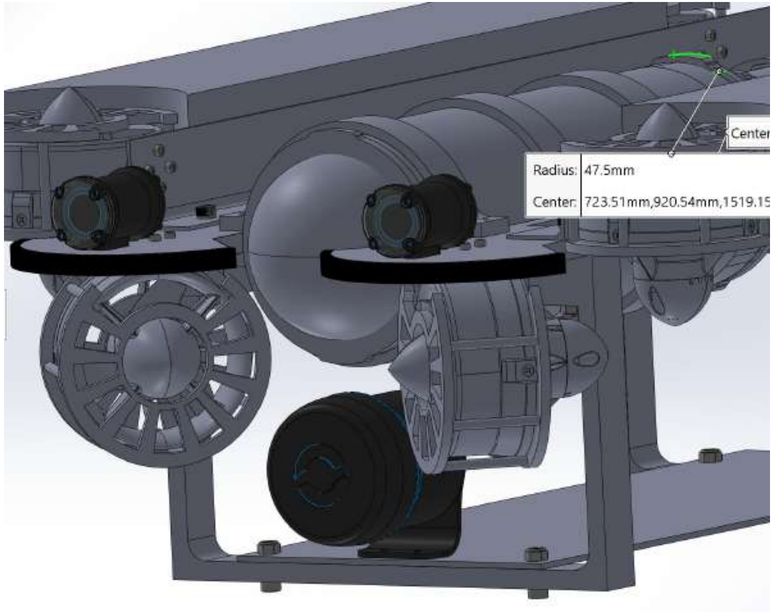


Connector Space Chassis Modification allows for shorter wire lengths

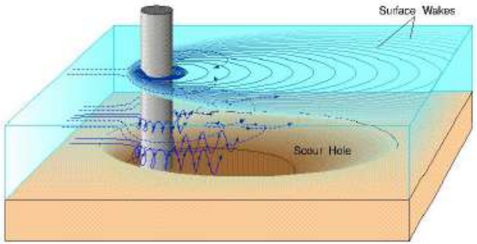


- Designed 1.25 inch gap designed between enclosures to accommodate connections into electronics enclosure (adjustable)

Chassis Modifications - Modular Platform



- Modular bottom platform mitigating thruster impacts, accommodating desired sensors / effects (lights, cameras, etc.)

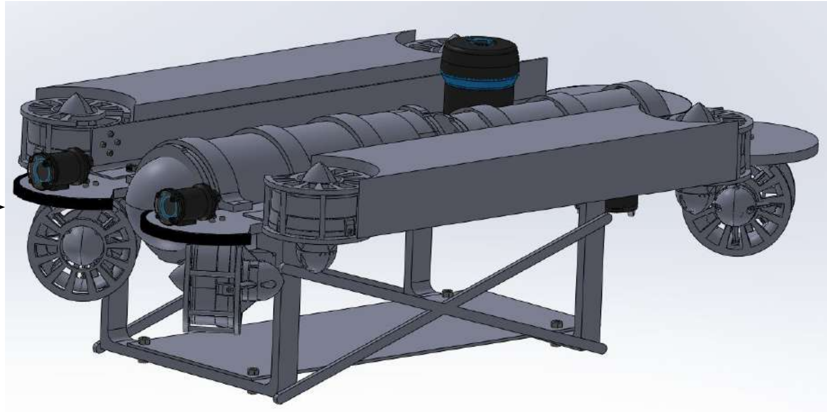


(b)

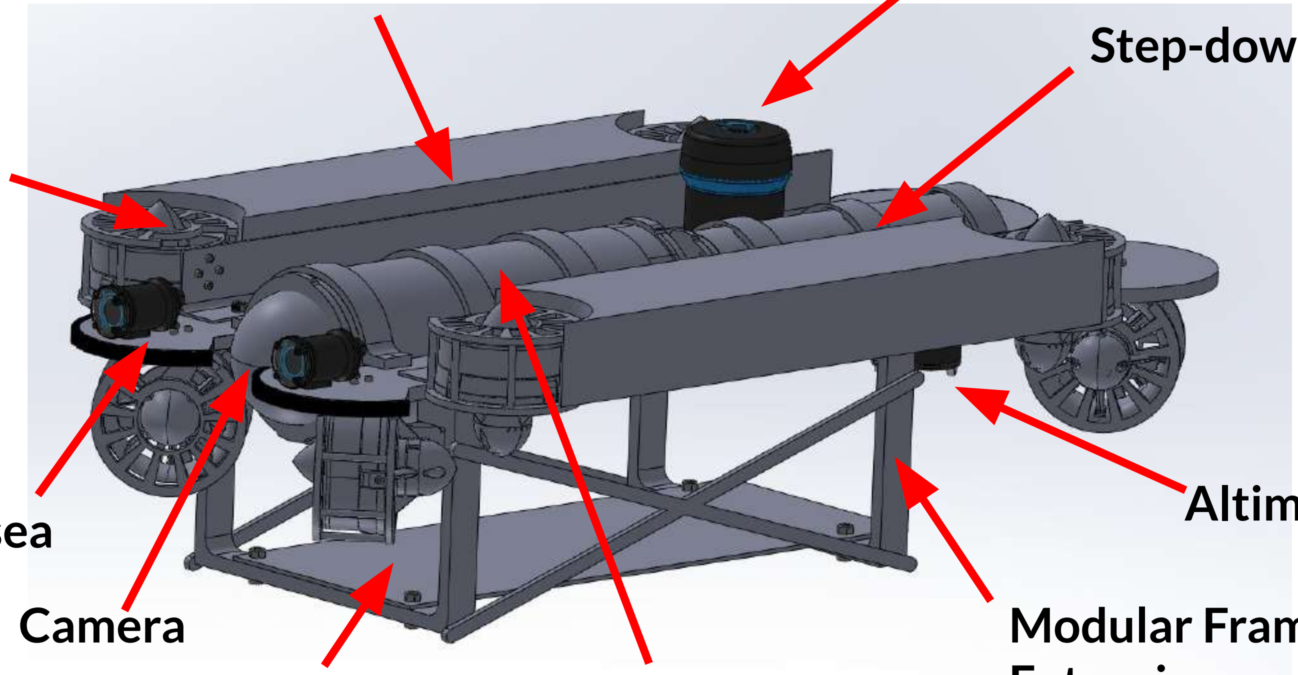
ROV Objectives Defined the Design

- Reliably **locate and navigate** ROV
  - Crucial for remote operation
  - Prevent collisions and/or entanglement
- Visually **identify** seabed erosion, cracks
  - Obtain images for structural analysis of **individual** wind turbine/asset foundations

Off-The-Shelf BlueROV2 Heavy allows for Easy Modification to fit REMORA Objectives

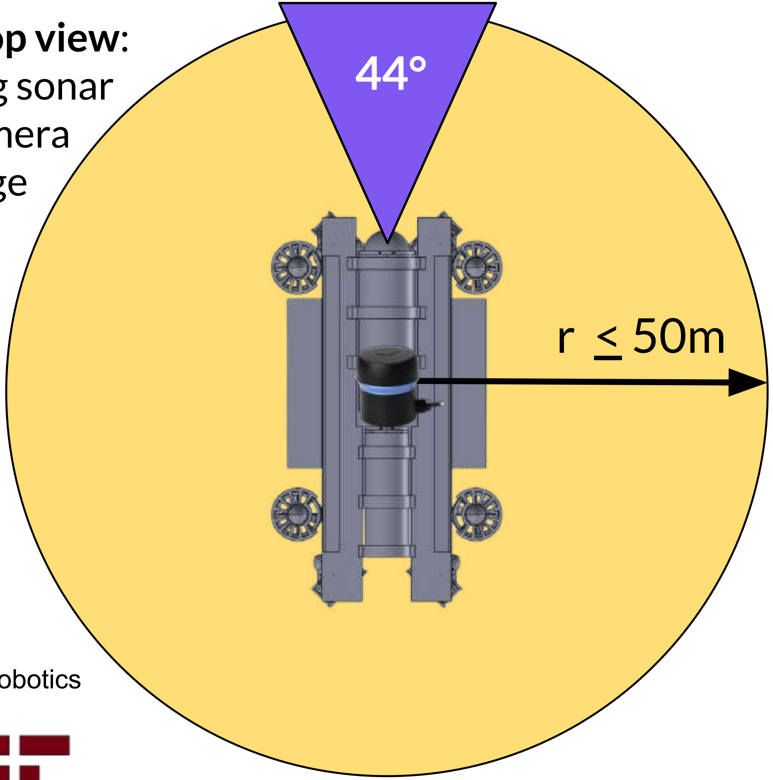


Re-designed BlueROV2

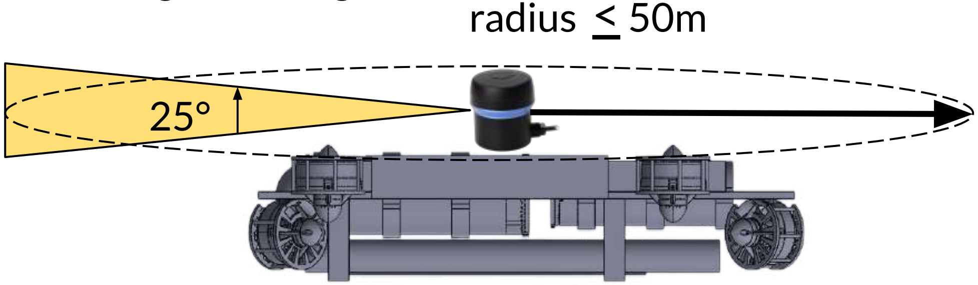


Sonar and Cameras provide full 360˚ Coverage in X-Y Axis

ROV Top view:
Imaging sonar and camera coverage



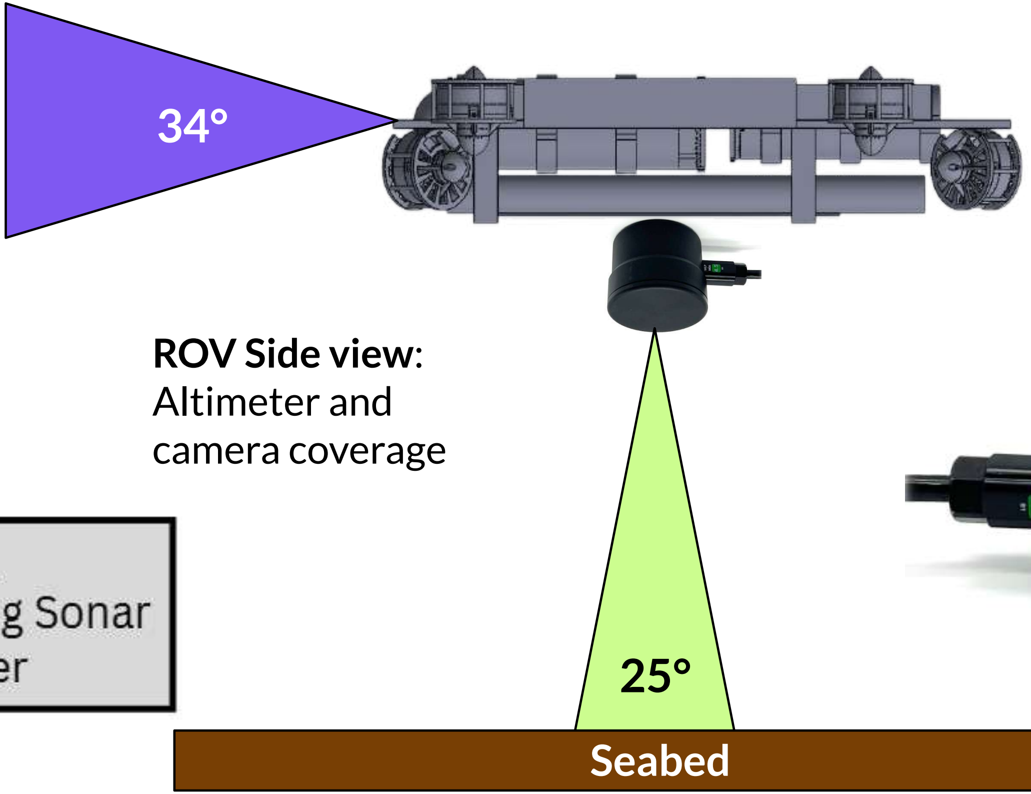
ROV Side view:
Imaging sonar angle coverage



Camera
Scanning Sonar
Altimeter

[3] BlueRobotics

Altimeter Sonar Provides Z-Axis Localization

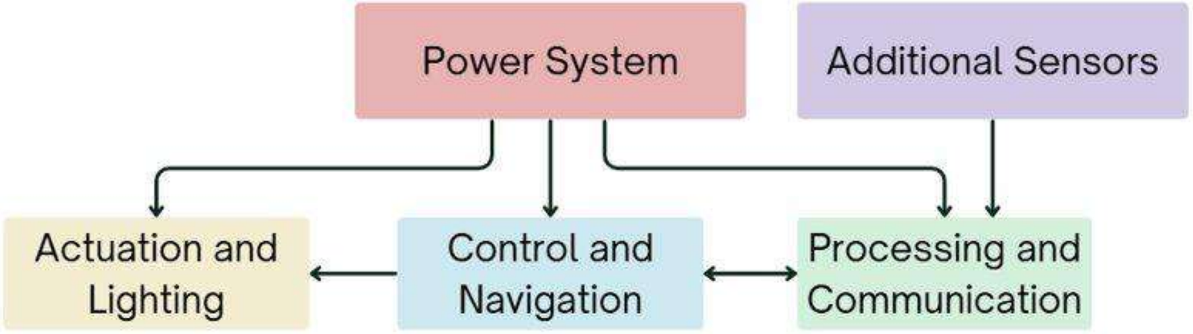


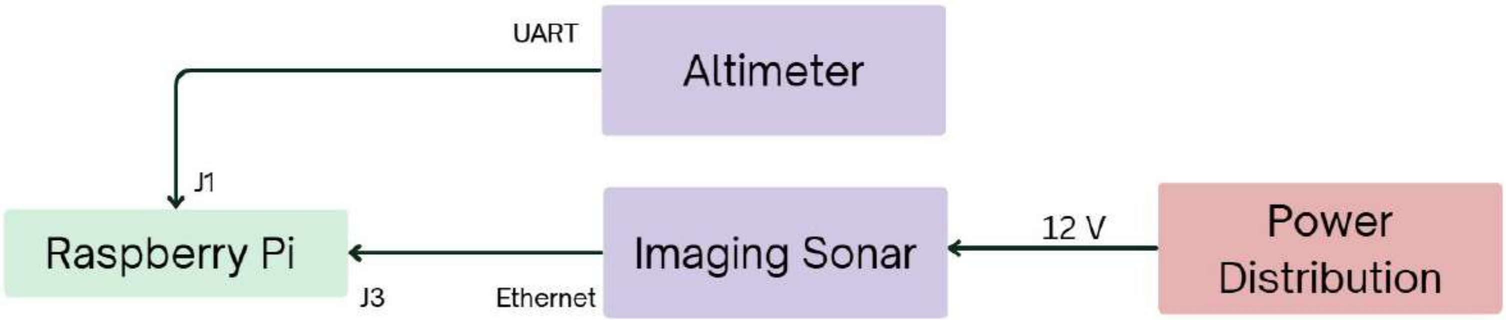
ROV Side view:
Altimeter and
camera coverage

Camera
Scanning Sonar
Altimeter

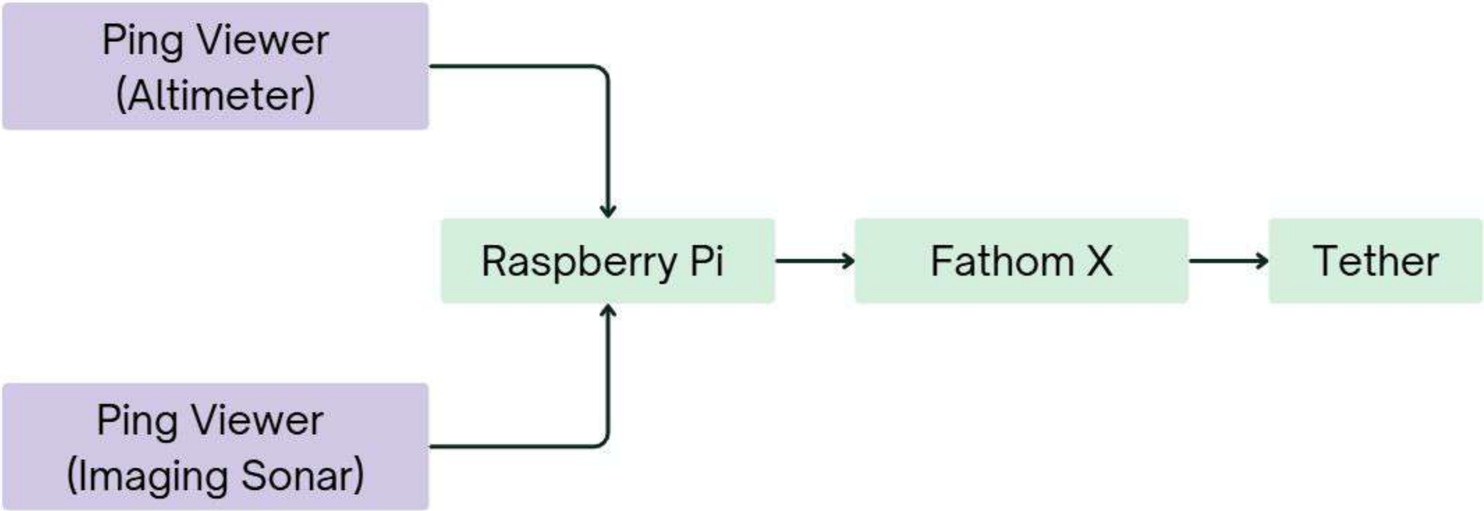
[3] BlueRobotics

Sensors Integration





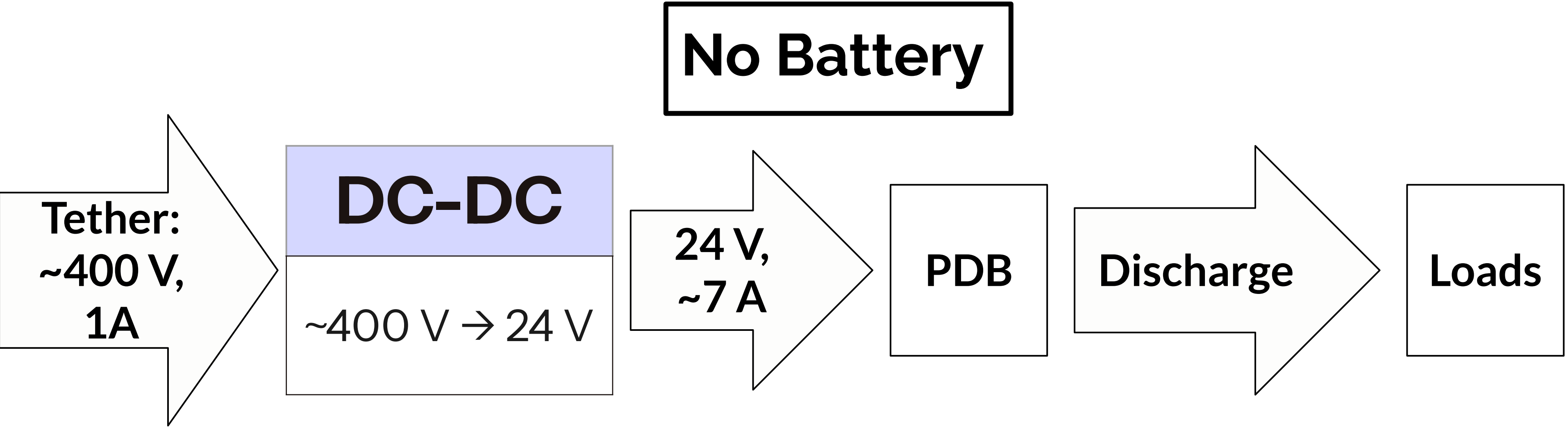
Sensors Integration: Data Transmission



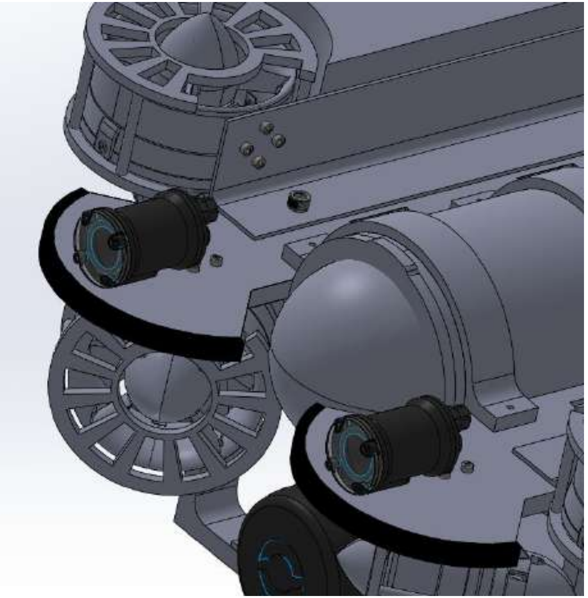
Power Relevant Design Requirements

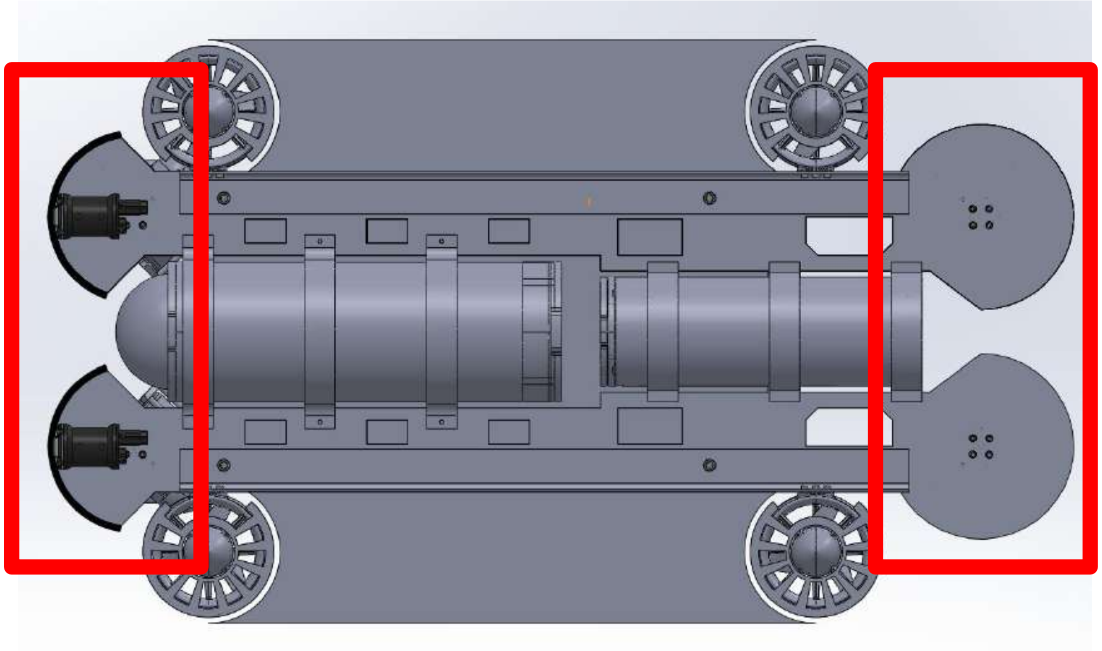
Requirement	Derivation	Numerical Value
Must be able to cover full foundation area	100 m tall, 5 m radius cylindrical foundation	3944 ± 95 m
Reasonable Foundation Inspection Time	2 weeks total mission time	$\leq$4 hours
Must Power Energy Consuming Electronics	Camera Specs	>= 99 ± 15 W / 72 ± 12 Wh

No Battery Architecture Overview



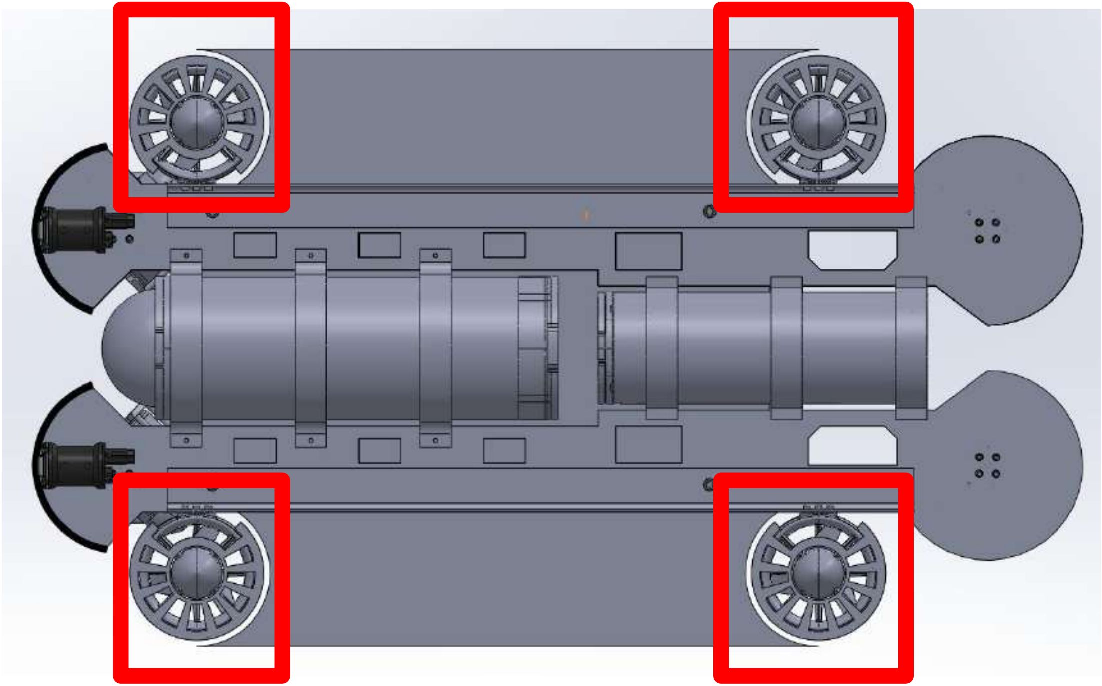
Impact Resistance Mitigation via Rubber Bumpers



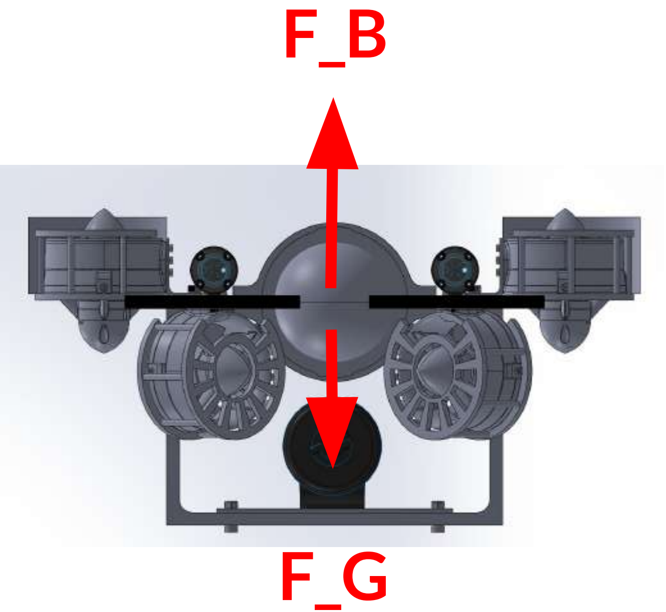


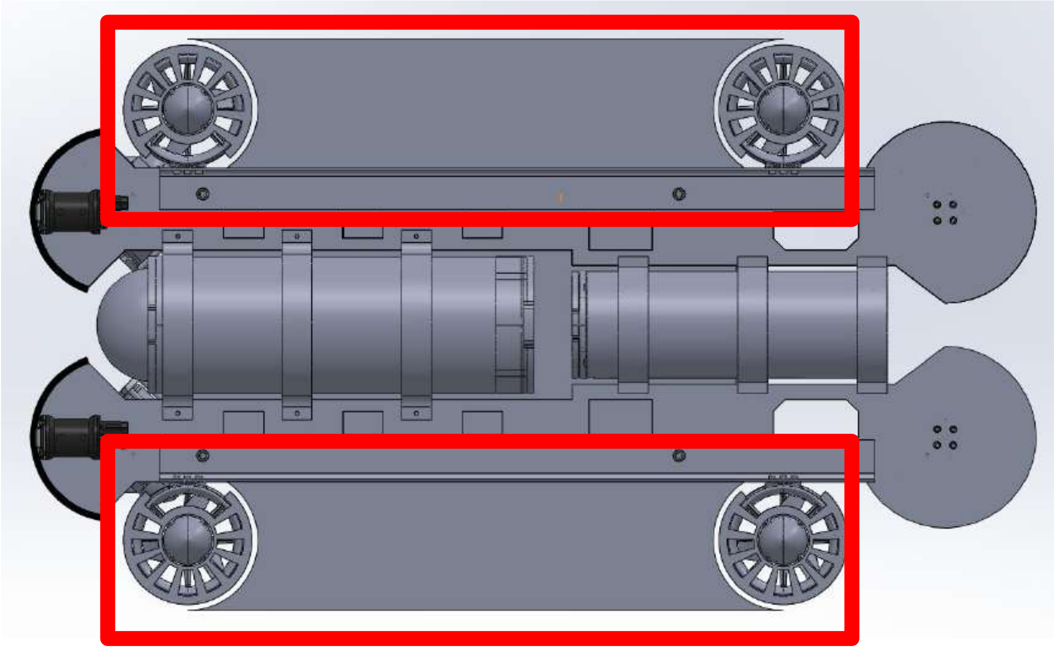
Thruster Configuration Enables Sufficient Thrust

- **29.1 lbf** forward/reverse/sideways
- **46.4 lbf** upwards
- **36.0 lbf** downwards

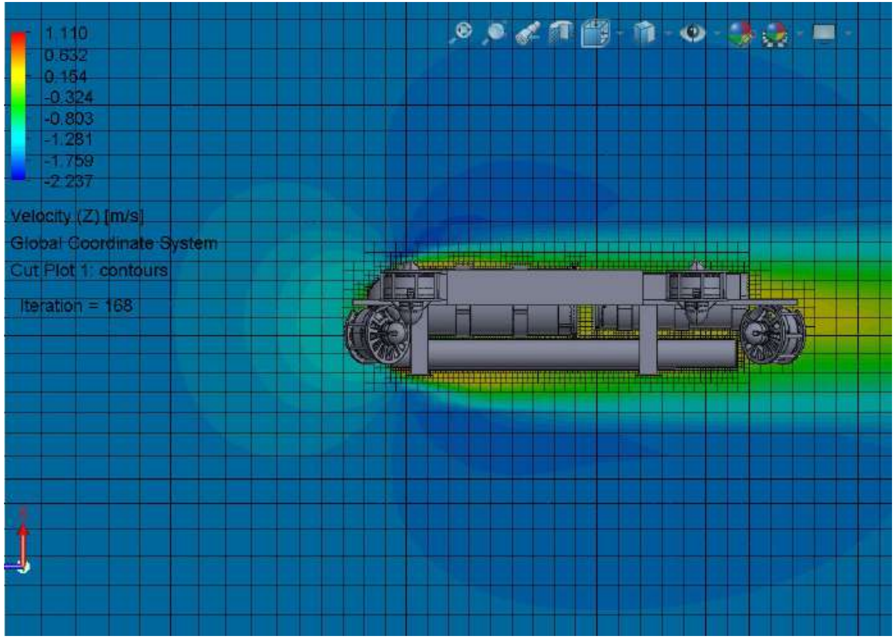


Reinforced In-Water Stability via Buoyancy Foam





Hydrodynamic Analysis show Minimal Drag Effects



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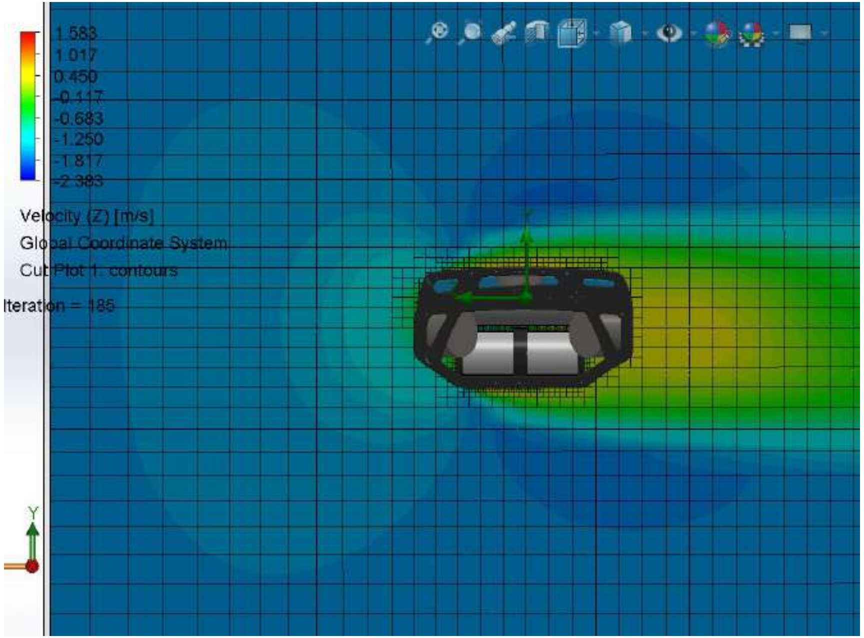

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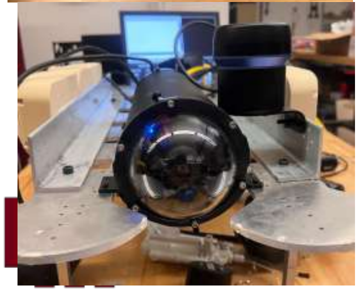


Manufacturing

- Electronics received 4/1
- Severe difficulties with obtaining manufacturing resources
- Pivoted to SendCutSend parts 4/10
- Parts received 4/17
- Manufacturing and major component assembly finished 4/23

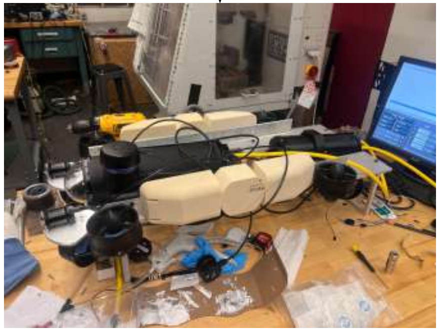
Testing

- ROV completed 4/24

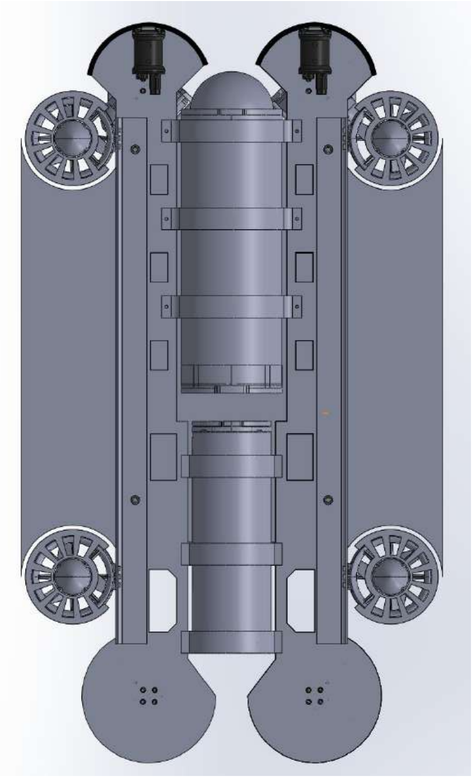
-Initial ballast tuning 4/24

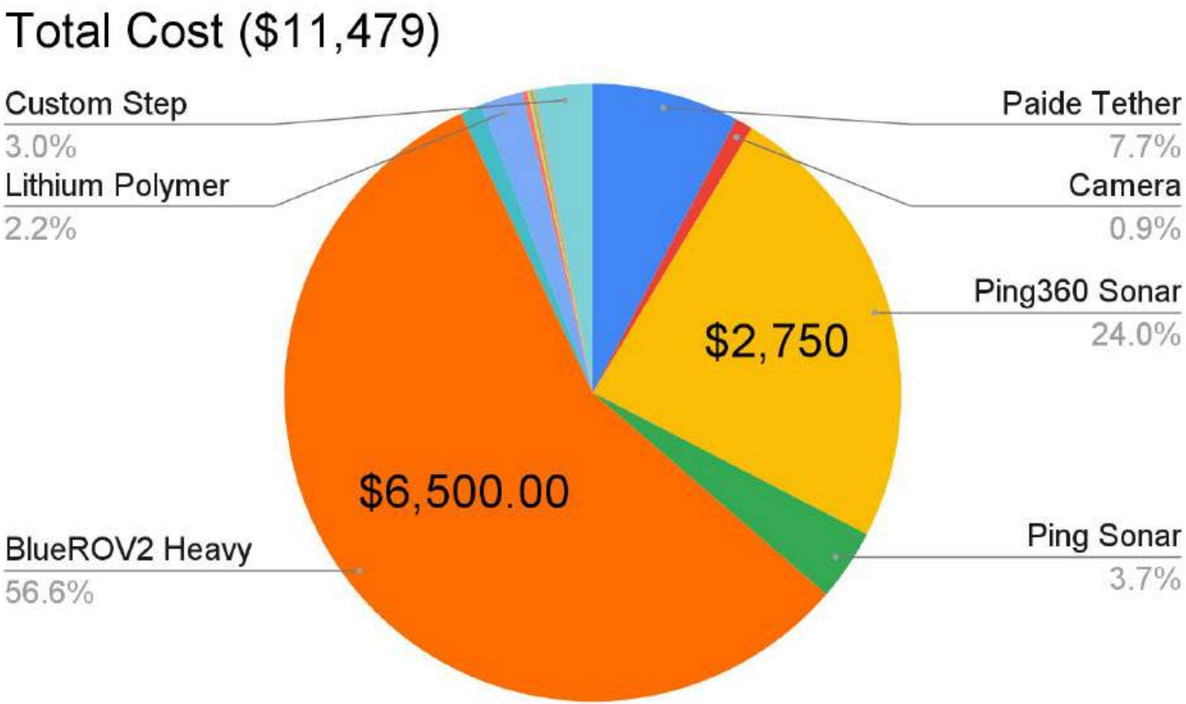
-Dry testing 4/25

-Wet testing 4/29



Cost Breakdown

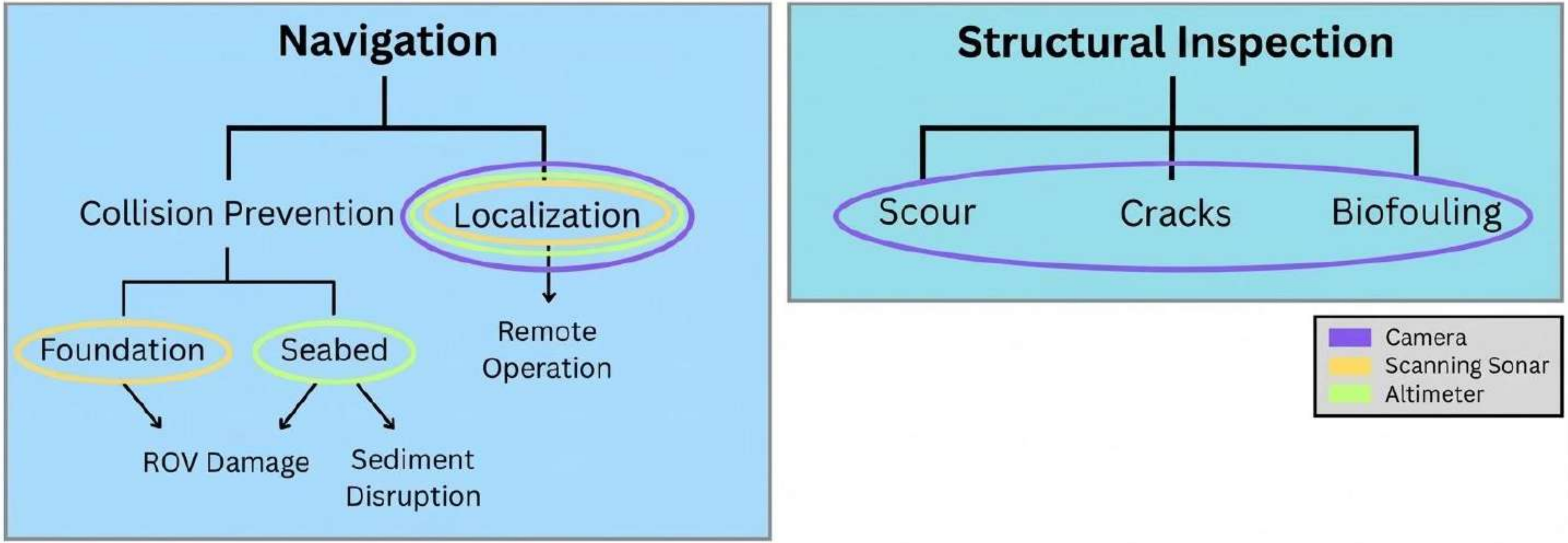




High Level Sensor Design Requirements

- **Localization and Navigation**
  - Avoid obstacles (manmade and natural)
  - No collisions with foundation or seabed
  - Stay 1.5 m above seabed to prevent uplifting sediment
- **Structural Inspection**
  - View 0.2 mm cracks on foundation from 0.5 m away
  - Identify scour near seabed

Sensor Design Requirements Flowchart



Sensors Relevant Design Requirements

Requirement	Justification	Value
Height from Seabed	Must not lift sediments during navigation (Critical Erosion Velocity is 1.0 m/s)	$\geq$ 1.5 meters [1] Avendaño 2024
Small Crack Visibility	Must detect average steel crack (0.2 mm) at 0.5 meters distance.	$\frac{1.5W_{sensor}}{X_{pix} f_D} \leq 0.2\ \mathrm{mm}$ [2] Verheij 1983
Foundation Detection	ROV is deployed <= 75 m from foundation	$\leq$ 75 m

Sensor Selection

Sensor	Capabilities	Specs
Ping360 Scanning Imaging Sonar	• 2D, multi-beam sonar • Allows for obstacle avoidance + target identification • Necessary under poor visibility conditions	Range: 0.75 - 50 m Resolution: 0.08%
Ping Sonar Altimeter	• Single-beam sensor (single distance reading) • Crucial for avoiding seabed collision + lifting sediment	Range: 0.3 – 100 m Resolution: 0.5%
Raspberry Pi HQ Camera	• High resolution sensor compatible with long lenses. • For remotely operated ROV navigation and imaging for structural defects	12.3 MP, f = 12 mm

Power Overview - ROV

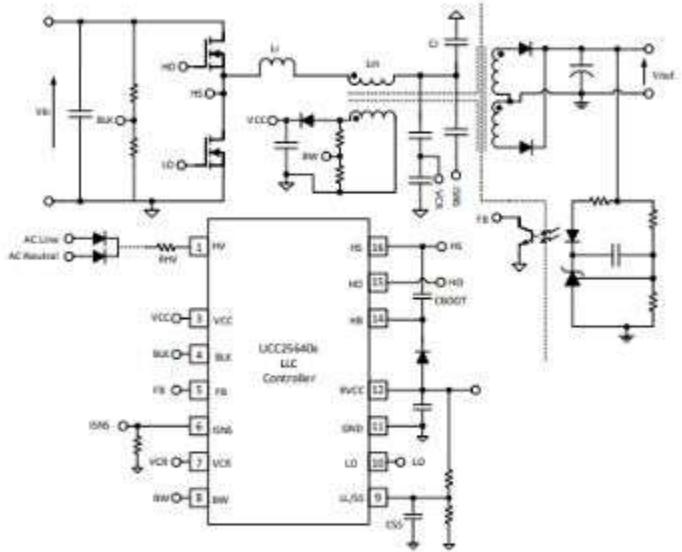
- **Sensors, lights, and propellers** on the ROV needs to be fully powered for the whole duration of the mission through the tether
- Must be **continuously powered** through USV power source connected by the tether

Battery vs. No battery

Battery	No battery
• Increased ROV recoverability if cable is cut, but have determined that it's not a hard requirement	• No battery maintenance • Less complexity/failure modes • Lower cost

Custom Step Down (ROV)

Estimated cost ~$300



- Half-bridge LLC converter
  400V step-down, max 500W
- Smaller form-factor, cheaper than off-the-shelf

- Requires water-proofing

ROV Chassis Objectives

- Must not permanently deform under all expected loads
  - Crucial for consistent operations, minimizing chance of vehicle loss
- Visually **identify** seabed erosion, cracks
  - Obtain images for structural analysis of **individual** wind turbine foundations

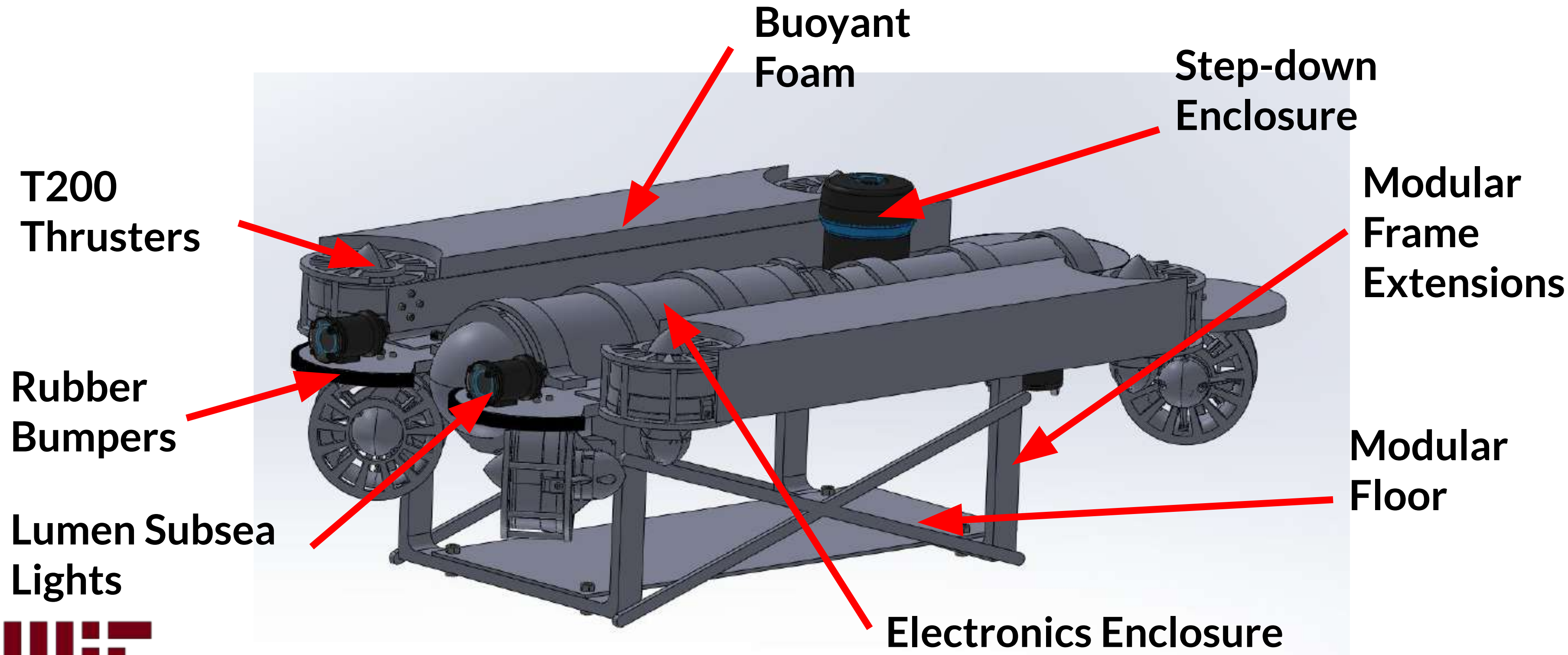
Requirements Check - Chassis (1/2)

Impact Resistance	Must be rated to withstand impacts with debris and during docking	Collision at maximum speed into seafloor	Need to ensure that there will not be a mission critical failure in case of collision
Debris Mitigation Resistance	Must include mitigations against prop fouling	N/A	Loss of propeller control could lead to mission failure
Cost	Must be less than provided budget by 2.013 for prototype	<$25,000	Cannot build if greater

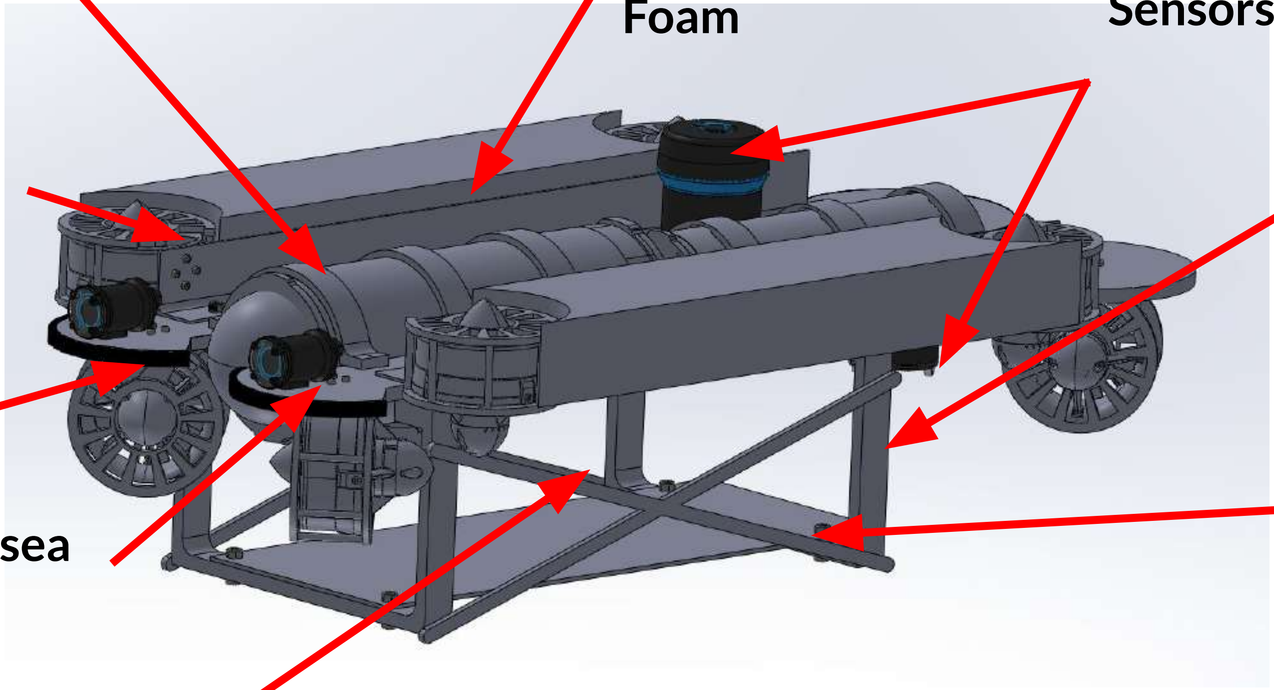
Requirements Check - Chassis (2/2)

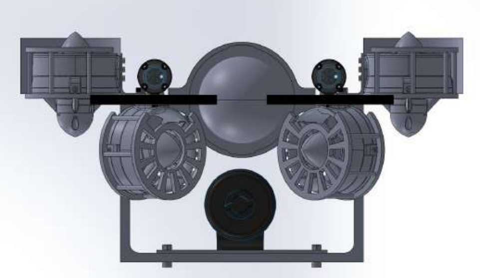
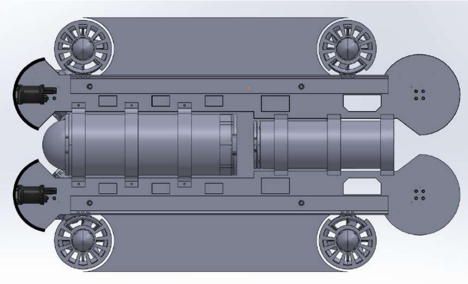
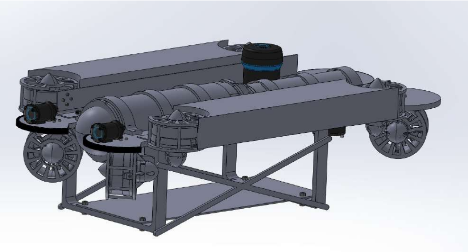
Weight	Must be light enough to be carried by USV	<300 kg	Hits the US Navy Lightweight UUV Category Range.
Stability	Must be statically stable in undersea and surface conditions	N/A	If flips over in any expected dynamic condition, could lead to vessel loss
Propulsion	Must have on-board means of propulsion	N/A	If no means of propulsion, no means of control or reliable data acquisition

Chassis Review



Modular Chassis Overview

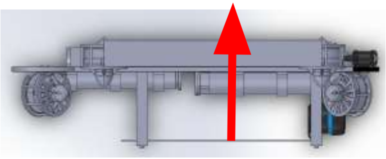
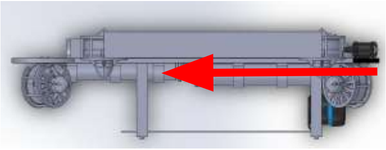
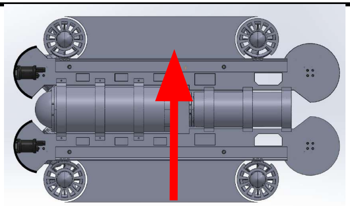


Chassis Modifications

- Modifications to BlueROV2 Heavy
- Modification to underwater “tray” design to accommodate easier height changes
- Thruster guards for protection against underwater debris
- Retains electronic and battery enclosure architecture as-is
- Retains 6-DOF
- All fasteners added to CAD (all M3 / M6 bolts)

Load Cases

Impact - Bottom	Slamming into hard surface at 1 m/s		Hand calculation done for middle of skid (weakest point)
Impact - Front	Slamming into hard surface at 1.5 m/s		If flips over in any expected dynamic condition, could lead to vessel loss
Impact - Side	Slamming into hard surface at 1.5 m/s		If no means of propulsion, no means of control or reliable data acquisition

Hydrodynamic Analysis

- Testing parameters:
  - v = 1.66 m/s (1.5 m/s operating speed + 3% Beaufort 3 surface current)
  - $\rho_{seawater}$ = 1025 kg/m^3
- BlueROV2 Base (A = 0.063 m^2): <u>0.42</u>
- Modified Chassis (A = 0.050 m^2): <u>0.44</u>
- Negligible increase in drag from additions

Requirements Check - Sensing

Requirement	Derivation	Value
Min Height from Seabed	Must not lift sediments during navigation	5 meters
Camera Cone	Must photograph perimeter in reasonable time	50° (V & H)
Camera Quality	Better crack detection	8 MP
Monopile Detection	Should have ROV dropped <= 50 m from monopile	50 m
Water Quality	Should measure various signs of biodiversity	RDO, pH, turbidity, ammonium

Requirements Check - Operations

Requirement	Derivation	Numerical Value
Distance travelled per foundation	Must survey 100 m tall, 5 m radius cylindrical foundation	3944 ± 95 m
Time per foundation	Should accommodate 2 weeks total mission time	<2 hours
Power/Energy Consumption	Should deliver adequate power for sensors	>= 99 ± 15 W / 72 ± 12 Wh

BOM - ROV - Structural / Electrical

Object	Qty	Unit Cost	Total Cost	Description	Link
BlueROV2 Heavy	1	$9240.00	$9240.00	Base Underwater robot	BlueROV2 Heavy Configuration for Stability and Manueverability
HDPE Stock	2	$64.56	$129.12	Additional Stock for Chassis Modifications	Marine-Grade Moisture-Resistant HDPE Sheet, 24" x 48", 1/4" Thick | McMaster-Carr

BOM - ROV - Sensing

Object	Qty	Unit Cost	Total Cost	Description	Link
Paide Tether	1	$885.00	$885.00	Powered Tether	Coupling+Docking - Google Drive
Camera	1	$104.95	$104.95	12MP Camera Upgrade	Raspberry Pi High Quality HQ Camera [12MP]
16mm Lens	1	$77.50	$77.50	Camera Lens	16mm Telephoto Lens
Ping360 Sonar	1	$2750	$2750	Imaging Sonar	Ping360 Scanning Imaging Sonar
Ping Sonar	1	$430	$430	Altimeter	Ping Sonar Altimeter and Echosounder

Power BOM

Object	Qty	Cost	Description	Link
Custom Converter	1	~$500	Voltage step down (400V to 24V)	2